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Scheduling aerial resource operations for the extinction of large-scale wildfires☆

Nina Skorin-Kapov [a,*], Luka Mesarić [b], Fernando Pereñíguez García [a], Lea Skorin-Kapov [b]

[a] Centro Universitario de la Defensa, Universidad Politécnica de Cartagena, C/ Coronel López Peña S/N, Base Aérea de San Javier, Santiago de La Ribera, 30720 Murcia, España

[b] University of Zagreb, Faculty of Electrical Engineering and Computing, Zagreb, Croatia

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ABSTRACT

The significant increase in large-scale wildfire events in recent decades, caused primarily by climate change, has resulted in a growing number of aerial resources being used in suppression efforts. Present-day management lacks efficient and scalable algorithms for complex aerial resource allocation and scheduling for the extinction of such fires, which is crucial to ensuring safety while maximizing the efficiency of operations. In this work, we present a Mixed Integer Linear Programming (MILP) optimization model tailored to large-scale wildfires for the daily scheduling of aerial operations. The main objective is to achieve a prioritized target water flow over all areas of operation and all time periods. Minimal target completion across individual areas and time periods and total water output are also maximized as secondary and ternary objectives, respectively. An efficient and scalable multi-start heuristic, combining a randomized greedy approach with simulated annealing employing large neighborhood search techniques, is proposed for larger instances. A diverse set of problem instances is generated with varying sizes and extinction strategies to test the approaches. Results indicate that the heuristic can achieve (near)-optimal solutions for smaller instances solvable by the MILP, and gives solutions approaching target water flows for larger problem sizes. The algorithm is parallelizable and has been shown to give promising results in a small number of iterations, making it applicable for both night-before planning and, more time-sensitive, early-morning scheduling.

1. Introduction

In recent decades, evidence has pointed to significant shifts in fire regimes worldwide with respect to frequency, size and intensity [1]. Hot, dry and windy weather conditions, which are conducive to wildfires, have become more frequent in several regions and are expected to keep increasing with higher levels of global warming [2]. Such weather is the cause of longer fire seasons, more intense mega-fires and fires in previously lower-risk regions. This, in turn, changes the biomass and releases more climate-changing carbon and other greenhouse gasses into the atmosphere, resulting in a positive feedback loop. Although a declining trend in the overall frequency of fires can be observed since the 1990s, particularly of smaller intentional landscape-related events, there is an increased occurrence of mega-fires (wildfires which burn more than 100k acres) and burned areas [1,3,4]. The year 2022 has been particularly devastating in Europe, where both fire frequency and area burned are 3-times higher than the annual average over the past 15 years [5]. Note that most damage caused by fires is due to extreme wildfire events, which only account for about 2% of the total number of fires [4]. It is projected that changing climate conditions, as well as land-use change, will result in a global increase in the likelihood of extreme wildfire events of up to 30% by 2050, and even up to 50% by the turn of the century [1].

Growing concerns regarding the ecological, land resource and socio-economic impacts of such changing fire regimes require that management strategies and programs be adapted accordingly [4]. In addition to increased prevention and awareness, which are key policy challenges, effective reaction and suppression mechanisms to deal with large-scale wildfires are imperative to reducing casualties and other damages. The extinction of wildfires involves several important resource allocation and scheduling decisions regarding personnel, ground and aerial resources. Wildfires of smaller scale are tackled promptly, with real-time dynamic resource scheduling of a small set of resources. On the other hand, large-scale fires can last several days, even weeks, employing a larger number of resources requiring more complex protocols and

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* Corresponding author.
E-mail address: nina.skorinkapov@cud.upct.es (N. Skorin-Kapov).

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planning models in order to maximize suppression efforts and ensure required safety measures. Such decision-making typically takes place daily, or even multiple times a day, to adapt to changing fire conditions, thus, aimed at shortening the duration of the fire and minimizing damage to the environment.

To extinguish or control a fire, either fuel, heat, or oxygen must be removed. Aerial resources are frequently employed to help combat wildfires by supporting firefighting crews on the ground who ultimately extinguish the fire [6]. Airplanes and helicopters drop water or chemical fire retardants near the edge of the burning fire, lowering the air temperature and sometimes creating a barrier between the wildfire and available fuels, referred to as a firebreak [7].[1] This allows ground crews to come closer to the fire and extinguish it or improve the created firebreak by removing fuel (vegetation or other combustible material) with hand tools. While smaller-scale fires may involve only a few aerial resources, mega-fires can involve 10, 20, or even more. As an extreme case, in a mega-fire which occurred in the province of Málaga, Spain, in September 2021, 51 airplanes and helicopters were engaged in aerial firefighting in a single day to assist almost a thousand firefighters on the ground [8].

Given the increasing number of aerial resources used in wildfire suppression, the planning, operations and logistics associated with them requires dedicated emergency resource management and careful coordination by the aerial coordinator [9]. Furthermore, the management of aerial operations must be centralized for the entire fire so as to increase the safety and efficiency of operations. Present-day management and coordination lack efficient and scalable algorithms to aid such complex decision making. While some works exist in the literature to deal with smaller-scale fires, they are not directly applicable to mega-fires, not only due to scalability issues, but also because restrictions, assumptions, and objectives related to mega-fires can differ from smaller-scale fires. Consequently, models should be adapted to effectively deal with different fire regimes.

In this paper, we present a novel optimization model tailored to large-scale wildfires for the daily scheduling of aerial operations by temporally allocating aerial resources to areas of operation. The main objective is to ensure target water flows in all areas of operation and continuity over time in accordance with a given attack strategy, ensuring prioritized non-negligence of fire areas and time periods. The model also aims to maximize minimal target completion and total suppression efforts. We develop a Mixed Integer Linear Program (MILP) model, which can provide optimal solutions using lexicographic optimization for smaller-sized instances of up to 15 aerial resources and 3 areas of operation. For larger instances, we propose a fast and effective multi-start heuristic approach using a randomized greedy (RG) algorithm coupled with simulated annealing (SA) employing large neighborhood search (LNS) techniques. The algorithm is referred to as Iterative RG-SA (It-RG-SA).

To test the approaches, we create a diverse set of 320 tests instances based on current legislation and data related to aerial firefighting in Spain which can serve as benchmark instances for future works. Results show that the heuristic algorithm obtains optimal or near-optimal solutions for smaller instances solvable by the MILP. For larger instances, where the optimal solution is unknown, the heuristic approach is able to find solutions which approach the target water flows in a reasonable time for most problem sizes. The multi-start aspect allows for parallelization and tuning of the execution time to adapt to different time restrictions or application scopes.

The rest of this paper is organized as follows. Section 2 provides a literature review, followed by the problem definition and MILP optimization model in Section 3. Section 4 describes the proposed heuristic and Section 5 presents computational results. Finally, Section 6 concludes the paper and provides an outlook for future work.

[1] For simplicity, in this paper the term *water* also implies *fire retardants*, *foam fire suppressants*, or any other chemical dropped by aircraft.

2. Literature review

Forest fire management is a wide area of research encompassing several aspects, such as fire detection, fire localization, fire propagation modeling, fire suppression, victim localization, and evacuation planning. A recent survey of systematic approaches to managing forest fires can be found in [10]. Most work on optimizing resources in forest fire management is focused on operations research models for problems such as allocation of resources, spatial deployment and scheduling suppression operations. The most common objective is minimizing suppression costs and the potential damage caused by the fire. Early operational research works on forest fire management are outlined in [11] and more recently in [12]. The work in [13] identifies a broader set of objectives, decisions, and constraints that should be integrated into next generation operational research models, such as responder exposure and probability of success. Other non-market values, such as habitat, health, air and water quality, in the evaluation of wildfire management are discussed in [14]. A literature review on the development of operational models that emulate fire suppression as part of decision support systems is given in [15]. Optimization models for decision-making *after* the occurrence of a fire have also been proposed, such as early work on forest management decisions based on processing network models constructed for common forest behaviors including growth, fire, and pest infestation [16]. More recently, the work in [17] proposes a decision support model for determining an optimal post-fire harvesting plan, workforce assignment, and cash flow management for forestry companies.

We classify works related to resource allocation in wildfire suppression activities into two broad categories as in [15]: (i) *preparedness*: allocation and deployment prior to the occurrence of fires (e.g., annual budget plans and strategic deployment over a given territory according to forecasted events and/or risk analysis), and (ii) *response*: scheduling (dispatching) and allocating resources during actual suppression efforts in the presence of one or more fires. Several important decisions must be made in both cases, typically aimed at minimizing total costs and damage caused by the wildfire while maximizing the effectiveness of suppression efforts and wildfire containment. While costs play a more significant role in management issues prior to the actual occurrence of a fire, resource allocation and scheduling of firefighting operations during an actual fire typically aim to maximize suppression efforts and minimize casualties and damage [18].

Most work on *preparedness* in the context of resource allocation focuses on the allocation and strategic deployment of various land and aerial resources to minimize costs, while ensuring that sufficient resources are available and accessible to combat potential fires. In [19], a two-stage stochastic programming model is presented to determine the location of protection depots aimed at protecting valuable assets (e.g., hospitals, schools, communication towers) in anticipation of an escaped wildfire. In [20], a maximal coverage location model is proposed to optimize the location of fire-fighting vehicles covering a given area with the main objective of minimizing the potential damage caused by forest fires. In [21], a linear programming model is developed for the deployment of land resources so that any forest fire is attacked within a specified time limit, as well as a simulation module to decide how they should be reallocated as a particular fire develops. The work in [22] proposes stochastic models using the C+NVC (Cost plus Net Value Change) model, initially proposed in [23], for optimizing the location of wildfire-attack bases under uncertainty of fire occurrence, and the deployment of pre-firefighting resources under uncertainty of fire growth. In [24], a stochastic integer programming model is proposed for making strategic dozer deployment plans for wildfire initial response planning for a given fire season based on historical data. Works dedicated to the allocation of aerial resources can be found in [25,26]. In [25], a mathematical programming model is given for strategic management issues such as the acquisition and determination of initial bases and deployment strategies for airtankers to minimize annual costs while meeting daily demands. In [26], the deployment of aerial firefighting aircraft is optimized based on predictive fire risk estimations over a certain geographical area using heuristic approaches based on harmony search and greedy algorithms.

In the context of *response*, several works consider general emergency response operations, such as relief distribution, medical attention, and evacuation in the presence of fires and other disasters [27–29]. In [30], a heuristic is proposed for bushfire emergency evacuation modeled as a capacitated vehicle routing problem. For resource allocation relating to fire suppression, we can broadly divide research efforts into approaches that consider multiple simultaneous fires, typically of smaller size, or a single larger fire. One of the earliest works considering the allocation of resources in case of multiple fires is actually focused on aerial resources, where queuing theory and dynamic programming are applied to develop decision-making aids for the daily allocation of available helicopters to bases for their use in initial-attacks of multiple fires [31]. More recently, dynamic repositioning of aircraft to airbases and fires over the course of a day was studied in [32] based on an underlying Mixed Integer Linear Program (MILP), which minimizes expected damage over a lookahead horizon. Decisions depend on prevailing stochastic weather conditions, the current state of fires in the region, and the current assignment of aircraft to bases and fires. Other works mainly focus on land resources or both types of resources but without considering the specifics of aerial firefighting. Stochastic line production rates were included in an optimization-simulation model for allocating land resources to several simultaneous fires [33]. In [34], the scheduling of available fire fighting resources in the case of multiple forest fires is modeled as a job-scheduling problem with deteriorating job processing times and decreasing job values. A heuristic algorithm is proposed with the objective of maximizing the total remaining value of the burnt areas. In [35], a minimal stochastic process representing wildfire progression is proposed and coupled with a series of response models considering trade-offs in response strength and timing and the division of limited resources for multiple competing suppression efforts.

Response models to optimize resources for individual fires include the following. The work in [36] presents an optimization model for resource management to minimize the value of burnt surfaces using simulation models for the prediction of fire behavior, resource combat, and meteorological conditions. An ILP model is proposed in [37] to select a mix of fire-fighting resources to minimize the costs of suppression, preparation, and damages according to the C+NVC model. The study is focused mostly on land resources and associated costs. An iterated local search algorithm is proposed in [38] for optimizing the location of land, air and personnel resources to fight a fire by modeling the landscape as a gridded network and estimating fire propagation times between adjacent nodes. The main objective is to minimize the burned area, as well as the number of deployed resources. In [18], an integer two-stage stochastic goal programming model for wildfire response, including resource allocation and resident evacuation, is proposed to minimize the cost of operations, property loss, and the people at risk to be evacuated. In [39], a dynamic resource allocation system for wildfire suppression is proposed using Markov Decision Processes (MDP). Two approaches are proposed: the first is based on Monte Carlo Tree Search and the second approximately models the MDP with a MILP.

Most of the aforementioned models for wildfire response focus primarily on land resources. Only recently have specific requirements and constraints related to aerial firefighting operations been incorporated into resource allocation and scheduling problems for wildfire containment [40,41]. In [40], an ILP model is proposed for selecting and scheduling resources for the extinction of forest fires for both aerial and ground resources while adhering to Spanish regulations for rest periods of pilots and brigades. The objective is to minimize the costs incurred for wildfire containment in terms of resource usage and affected land while ensuring a minimal required number of resources active over time. A second objective function is proposed to maximize total suppression efforts over time. In [41], two ILP models are proposed for the dedicated optimization of aerial resources. The first deals with the allocation of aerial resources to flight routes (fire fronts and recharge points) and the second deals with the assignment of refueling points. Both models assume the scheduling of aerial resources is predetermined and only consider the current time period. The objective of the first model is to maximize the total drops while minimizing the lack of water at each front relative to initially assigned values. The second model aims to minimize refueling time, subject to available fuel and distances.

In this paper, we propose a unified model for full day scheduling and assignment of aerial resources to areas of operation in a large-scale fire. While the models in [40,41] optimize aerial resources in wildfire suppression, they consider individual subproblems solved separately, and solution approaches are based solely on ILP formulations which do not scale well to mega-fires with a large number of aerial resources for full day scheduling. Since large-scale fires can last several days, and legislation limits the number and duration of daily flights, it is important to consider the entire daylight period to avoid intervals with lack of water, even if the tentative schedule is adapted as the fire evolves over the course of the day. In this paper, we incorporate assumptions related to large-scale fire suppression efforts to develop a consolidated MILP model which can find optimal solutions for the integrated problem for full day planning of up to 15 aerial resources and 3 fire areas. As opposed to commercial flight scheduling models such as those in [42,43] where minimizing costs is the main objective, flight scheduling for aerial firefighting in practice is aimed at maximizing suppression efforts. Consequently, our model aims to achieve a desired water target in each area of operation and each time period, to ensure not only prioritized non-negligence over fire areas but also prioritized non-negligence over time. Minimal target completion and total water output are also maximized as secondary and ternary objectives, respectively. For larger instances and/or time-sensitive application scopes, we develop an efficient heuristic approach to find sub-optimal solutions in reduced time.

3. Scheduling aerial resource operations: problem definition and MILP model

3.1. Context and problem statement

In practice, upon detection of a fire, a small set of resources (typically 3 or 4) is automatically dispatched to the fire area where the situation and potential for spreading is assessed. The fire manager or incident commander, along with onsite personnel, estimates the resources needed and the extinction strategy to be followed. The number and type of aerial means required to comply with the extinction (attack) strategy are estimated *a priori* and can be influenced by atmospheric conditions, such as visibility and local wind conditions, obstacles present and support facilities [9]. Resources are then allocated according to availability, proximity and estimated requirements. In the case of large-scale fires, which can last several days, resource allocation will vary from day to day according to the current and estimated evolution of the fire. Currently, the estimation of required resources is a qualitative and subjective process relying primarily on the fire manager's experience and technical knowledge, although efforts are being made towards automated dynamic resource allocation for wildfire suppression [39].

Once available resources are allocated for the current or next day, associated firefighting operations must be scheduled in time and assigned to individual areas of the fire. The management of aerial operations should be centralized for the entire fire to reduce security risks and increase operation efficiency. For example, in Spain, the head of aerial operations and the aerial coordinator will assist the fire manager with large-scale fires. Regulations dictate that aerial resources are not permitted to fight fires during night-time hours [44]. Consequently, scheduling and planning operations for the next day in large-scale fires are typically done from sundown until midnight. The situation is then

Table 1
Summary of input parameters and decision variables.

Sets and indices	
$t \in \mathcal{T} = \{1, \dots, T\}$	Set of time slots
$k \in \mathcal{K}$	Set of aerial resources
$q \in \mathcal{Q}$	Set of aerial resource types
$m \in \mathcal{M}$	Set of areas of operation
Input Parameters	
$I_m \in \mathbb{R}^+$	Importance of area m
$V_{qk} \in \{0, 1\}$	Resource type: 1 if resource k is of type q, 0 otherwise
$T_k \in \mathbb{Z}^+$	Flight duration (in number of time slots) of a single flight for resource k
$R_k \in \mathbb{Z}^+$	Minimum rest time between flights (in number of time slots) for resource k
$N_k \in \mathbb{Z}^+$	Maximum number of flights in the entire time period for resource k
$P_k \in \mathbb{Z}^+$	Maximum number of continuous time slots the crew operating resource k can be physically present
$A_k^t \in \{0, 1\}$	Availability: 1 if resource k is available in time slot t, 0 otherwise
$B_{qm} \in \{0, 1\}$	Area restrictions: 1 if only resources of type q can operate in area m, 0 if unrestricted
$U_{km} \in \mathbb{Z}^+$	Number of pure transit time slots required for resource k to travel to/from area m
$W_m^t \in \mathbb{R}^+$	Minimum water target required at area m at time t (in liters)
$S_m \in \mathbb{Z}^+$	Maximum number of resources that can operate simultaneously in area m due to limited airspace
$C_k \in \mathbb{R}^+$	Water capacity of aerial resource k (in liters)
$D_{km}^t \in \mathbb{R}^+$	Average number of drops an aerial resource k can make in area m in time slot t if it is present in the area and fighting the fire for the full time slot
$E_{km}^t \in \mathbb{R}^+$	Average number of drops an aerial resource k can make in area m in time slot t in the time slot in which it is arriving or leaving the area
M	A positive large number
Decision Variables	
$c_{km}^t \in \{0, 1\}$	Takes value 1 if resource m initiates a flight towards area m in time slot t, 0 otherwise
Auxiliary variables:	
$s_{km}^t \in \{0, 1\}$	Takes value 1 if resource k is active (transiting towards/from or fighting the fire) in a flight associated with area m during time slot t, 0 otherwise
$e_{km}^t \in \{0, 1\}$	Takes value 1 if resource k arrives/leaves area m during time slot t, 0 otherwise
$x_{km}^t \in \{0, 1\}$	Takes value 1 if resource k is fighting the fire in area m during entire time slot t, 0 otherwise
$v_{qm}^t \in \{0, 1\}$	Takes value 1 if resources of type q are fighting the fire in area m in time slot t, 0 otherwise
$TC_{tm}^+, TC_{tm}^- \in \mathbb{R}^+$	Surplus (TC_{tm}^+) and deficit (TC_{tm}^-) of water target completion in time slot t at area m. Water target completion is then given by $TC_{tm} = TC_{tm}^+ - TC_{tm}^-$
$TC_{min}^+, TC_{min}^- \in \mathbb{R}^+$	Minimal water completion over all time slots and areas, $TC_{min} = TC_{min}^+ - TC_{min}^-$, where TC_{min}^+ is the positive component and TC_{min}^- is the negative component.
$WO \in \mathbb{R}^+$	Total water dropped (water output) over all areas and time slots

re-assessed in the early morning in case the evolution of the fire is not as expected.

The scheduling of aerial extinguishing operations is done in accordance with the extinction strategy designated by the fire manager while adhering to several safety, legislative, and temporal restrictions. The mobilization of resources should be such that continuity of aerial work is achieved over the *entire* firefighting time period, avoiding intervals with a lack of water flow. An intense and numerous attack at the beginning of the day can bring the fire temporarily under control, but it may be useless if there are no aerial resources available during the rest of the day [9]. Consequently, staggering resources to ensure a constant flow of water over all daylight hours, in addition to concentrating efforts according to different extinction strategies based on meteorological and topographical conditions, is crucial to maintaining the fire under control.

In this paper, we consider the full day scheduling and planning of aerial firefighting operations in large-scale wildfires. The main objective is to achieve a desired minimal target water flow in each time period and fire area according to the extinction strategy designated by the fire manager. Furthermore, the model aims to maximize minimal target completion to distribute resources more evenly and maximize total suppression efforts. The schedule is created for a single day, which is often a part of a multi-day fire suppression effort. Details of the optimization model are given in the next subsections.

3.2. Assumptions and input parameters

In this subsection, we describe the assumptions and associated input parameters to the problem, which are summarized in Table 1. We consider a daylight planning period divided into a set of discrete time slots $t \in \mathcal{T}$, $|\mathcal{T}| = T$, of fixed duration. A discrete scheduling model was chosen as it allows for easier and intuitive linear modeling of the constraints. In practice, minor variations in the schedule, such as a resource departing a few minutes later/earlier, would generally not cause significant changes in the objectives. Consequently, the inaccuracy inherent in discrete scheduling is assumed to be acceptable if a reasonably small time interval is applied.

Given is a set of aerial resources, $k \in \mathcal{K}$, with associated water capacities, C_k, and a set of geographical areas of operation of the fire, $m \in \mathcal{M}$. A large-scale wildfire is typically divided into multiple areas (drop zones and corresponding refill points) to which resources are assigned to operate during a flight. Note, areas in $\mathcal{M}$ refer only to the areas of the fire for which the fire manager requested air support. Areas assigned solely to ground forces or those with unacceptable flight conditions are not included in the model.

Some of the areas in $\mathcal{M}$ may be prioritized due to factors such as proximity to populated areas, described by parameter I_m indicating the importance of individual areas. In case of smaller fires, aircraft may work in different areas of the fire during a flight as the fire is being extinguished. However, we focus on large-scale wildfires, where it is common practice for aerial resources to spend a single flight fighting the fire in the same general area. Note, the number of working areas requiring air support is not necessarily related to the size of the burning area, as it depends on topography and meteorological conditions.

When multiple aerial resources are active in the same area at the same time, they fly in a carousel-like formation. The flight route is a circular path defined by the water refilling point, and the discharge point. To avoid overcrowding the air space, a limit S_m is set on the number of aerial resources that can fly at the same time in a particular area m. Furthermore, each resource is categorized by a type $q \in \mathcal{Q}$, and only resources of the same type can operate simultaneously in a single fire area at any given time. Namely, resources with different characteristics, e.g., rotary or fixed-wing aircraft, refill differently, and may have different flight paths, altitude intervals, and flight speeds. For example, amphibious airtankers refill by scooping from a sufficiently long body of water, whereas helicopters are often equipped with a bucket which they lower into the water while hovering at a low altitude. If both types of aircraft work simultaneously in the same area, airspace coordination is more complex and poses increased security risks. Consequently, we assume only resources of the same type can operate simultaneously at a fire area at any given time. The model is general, where resources can be categorized into broad types, e.g., fixed- vs. rotary-wing, or more specific types, such as fixed-wing land-based or amphibious, light, medium and heavy helicopters, etc., according to safety and airspace coordination restrictions.

Some areas may only permit a certain type of resource to operate, described by input parameter B_{qm}. Namely, helicopters can refill from a small body of water (e.g., a narrow river, a small lake, swimming pools, etc.), whereas amphibious airplanes cannot. If an area has only such smaller water points in its vicinity, airplanes will have very limited usability as they will have to make long flights in order to refill at a further, larger water point. Additionally, if the terrain is particularly difficult, it might be inaccessible to airplanes.

Each area of operation may have one or more dedicated water points where aircraft can refill their water tanks. Assignment of water points in each area to each resource is assumed to be predetermined, mostly based on the resource type, and is not a part of the schedule. This assignment is reflected in a parameter related to the estimated number of drops per time slot and area for each resource (D^t_{km}) assuming the resource is already present and operating in the area. The number of drops may differ between areas due to the locations of refill points and may differ between resources, particularly if they are of a different type. In practice, a group of resources of the same type (particularly fixed-winged aircraft) simultaneously working in a specific area will typically fly in carousel refilling at the same water point. The number of drops of each resource or resource type in a given area may differ over time due to meteorological conditions affecting flights and refill constraints. For example, stronger winds may force an aircraft to fly slower or take a different, longer flight path. Note, parameter D^t_{km} representing the estimated number of drops per area and time slot is given as a real number, and not an integer, because it represents an estimated average value.

Each aerial resource has a maximum flight duration in terms of time slots (T_k) and a maximum number of flights it can perform daily (N_k) according to legislation. For example, in Spain, airtankers can perform up to two flights a day, with a maximum flight duration of 4 h (civilian) and 4 h 30 min (military), while helicopters are restricted to four flights of a maximal duration 2 h each. A minimum resting period (R_k) is also prescribed by legislation, depending on the duration of the previous flight, and the aircraft type. In our model, we assume all flights are of maximal duration. While such an assumption would not be realistic for small fires, it is a reasonable assumption for multi-day suppression efforts for large-scale fires. While actual flight duration may vary slightly, by ensuring maximal flight time, we ensure that the legislation restriction will be met even if flights are shorter. Note, since we aim to maximize suppression efforts, and available resources tend to be more restricting than the number of resources operating simultaneously, most flights would be of maximal duration even if this assumption were omitted. However, we include it here as it significantly reduces model complexity, allowing for the solution of larger instances. Since we assume fixed maximal flight times, minimal resting periods are also fixed. Between two consecutive flights, the aircraft must be refueled, inspected and, if needed, minor repairs can be made during this time. We assume all resources can be refueled within the minimal resting period, which is usually the case in practice. The resting period after the final flight of the day can exceed daylight hours, but the flight time must be within the planning time window. While it could be possible for aircraft to perform transits at night-time, be it early in the morning or late in the evening, effectively extending the flight schedule, it is not considered in this model.

Further restrictions are imposed by limitations on the maximal time the crew can be physically present (P_k). For example, under Spanish regulation, a crew member must not be active for more than 12 consecutive hours to complete up to 8 h of flight [44]. We assume one flight crew will be assigned to one resource per day, as is done in practice, and consider a crew to always be available since the availability of aerial resources is typically the limiting factor. Consequently, we do not differentiate between the crew and aerial resource in this model.

Each resource may be stationed at a different location and will spend part of its flight time traveling to/from the assigned fire area. For example, helicopters can be stationed at a variety of locations and tend to be within 20 min of flight time from the fire [9]. Airplanes are required to be stationed at airbases or airports, and typically have longer transit times (e.g., 40 min). Given is the number of pure transit time slots it takes each resource to reach each area of the fire, U_{km}, during which no drops will be made. The remaining time needed to reach the area is reflected in a parameter describing the estimated number of drops in the time slot when the resource arrives/leaves the fire area, E^t_{km}. In these time slots, the aircraft spends part of the time traveling to/from the base and part of the time fighting the fire, so the number of drops is proportionally smaller compared to the drops in a firefighting time slot, D^t_{km}. Note, if an aircraft is stationed close to a fire front, it may not have transit slots, as the aircraft can reach the front within a single time slot.

We assume resources return to the same location between flights, considered to be their home base. This is typically done in practice to simplify refueling and ensure qualified mechanics are available for required revisions. While in some cases, such as engine failure or obstruction in the water tank, an aircraft may make an emergency landing at a different closer base, we do not consider this case here. In large-scale fires with a large number of aerial resources, aircraft will often be allocated from other provinces, or even countries, with several hours of initial transit. However, once the resource arrives at the general fire area, it will normally remain stationed at a closer base for posterior flights. In such cases, the aircraft may be unavailable for part of the day due to long-distance transit described by the availability parameter, A^t_k. Aircraft may also be unavailable for parts of the day due to their allocation to a different fire. Note, in cases where an aircraft is partially unavailable due to long-distance transit or other fire fighting operation, it will have fewer daily flights available for firefighting than those prescribed by regulations, reflected directly in parameter N_k.

Finally, we assume a given minimal water target (in liters), W^t_m, for each area in each time slot. The water target is meant to adhere to the general extinction strategy, while ensuring the minimal continuous water flow needed over time/areas to keep the fire under control, avoiding intervals with lack of water flow. According to factors such as the intensity and size of the fire, topography, and proximity to populated areas, some areas may require more water. Furthermore, the water target may vary over different time periods based on predicted weather conditions and/or different extinction strategies, such as concentrating a large number of resources at the start of the day or spreading them more uniformly over time.

3.3. Decision variables

Given the set of resources $\mathcal{K}$, areas of operation $\mathcal{M}$, time slots $\mathcal{T}$, and water targets W^t_m for each area and time slot, resources must be scheduled and allocated to areas of operation while adhering to all aforementioned safety, legislative, and temporal restrictions. Since flight duration is assumed to be fixed, and each resource operates in a single area during a single flight, the schedule is fully described by a set of binary decision variables c^t_{km} indicating whether a resource k takes off towards area m in time slot t. Additional auxiliary decision variables used to formulate the MILP model, such as those indicating the time slots in which resources are active, arriving/leaving or fighting the fire in a specific area, are described in Table 1. Included are also variables indicating the degree to which the water target is met in each area and time slot, referred to as target completion TC_{tm}. Note, target completion may be positive (surplus) or negative (deficit) so it is modeled with two auxiliary nonnegative variables: TC^+_{tm} (surplus) and TC^-_{tm} (deficit). If the water requirement is met, deficit TC^-_{tm} takes a value of 0, and TC^+_{tm} indicates the surplus. If water is lacking, TC^+_{tm} takes a value of 0, and TC^-_{tm} indicates the deficit. Similarly, TC_{min} indicates the minimal water target completion over all time slots and areas of operation and is modeled by auxiliary variables TC^+_{min} and TC^-_{min} indicating its positive and negative components, respectively. Finally, variable WO takes the value of the total water output dropped.

3.4. Objective

The objective is given as a set of three hierarchical objectives, ranked lexicographically, and given as:

$$\max f_1 = -\sum_{t\in\mathcal{T}}\sum_{m\in\mathcal{M}} I_m \cdot TC^-_{tm} \tag{1}$$

$$\max f_2 = TC^+_{min} - TC^-_{min} \tag{2}$$

$$\max f_3 = WO \tag{3}$$

The primary objective described in (1) is to achieve the desired water target in each time period and fire area. If this is not possible, the quantity of total deficit, or lack of water necessary to achieve the targets, is minimized, where the deficit at individual areas is weighted by the importance of the area. Note, f_1 is maximized in (1), as the total deficit is expressed as a negative value. In order to more evenly distribute the resources, the secondary objective, given in (2), maximizes the minimal target completion in any particular time slot and area. Namely, if the water target cannot be met, it is better to have minor deficits in multiple time slots/areas than one major deficit in a single area or time which may allow the fire to lose control. Furthermore, if the target can be met in all time slots and areas (all deficit variables are 0), it is desirable to more evenly distribute the surplus by maximizing the minimal surplus above the target over all time slots/areas. This also allows the model to set a lower water target ensuring a minimal continuous water flow, but still achieving a more evenly distributed water flow. Finally, as a third objective, the model maximizes the total suppression efforts in terms of total water dropped, as given in (3).

Note, the model does not explicitly consider suppression costs, as this is assumed to be considered in deployment decisions before the fire occurs and for the allocation of the resources available each day. In Spain, particularly for large scale fires, costs of suppression efforts typically do not enter the decision-making process in the response phase. However, the costs of potential damage caused by the fire can be incorporated into the water target parameter by, for example, prioritizing areas with higher damage potential.

The three aforementioned objectives are assumed to be hierarchical with decreasing priority. This translates to a lexicographic optimization problem whose objective can be denoted as:

$$lex\ max\ \mathcal{F} = \{f_1, f_2, f_3\} \tag{4}$$

The resulting problem can be solved using the lexicographic method which sequentially solves three single-objective problems, optimizing each of the three objectives individually. Once the optimal value for the first objective (f_1^*) is found, this value is added as a constraint ($f_1 \geq f_1^*$) in the next two problems. Analogously, the optimal value for the second objective (f_2^*) found by solving the second problem is added as a constraint ($f_2 \geq f_2^*$) in the third problem which optimizes f_3.

Alternatively, the problem could be solved as a single-objective problem using the weighted-sum method, where a set of weights is defined such that even the smallest change in a higher priority objective carries more weight than the largest change in a lower priority objective. The objective function is then given by:

$$\begin{aligned}\max\ \mathcal{F}_w = &-\alpha_1 \cdot \sum_{t\in\mathcal{T}}\sum_{m\in\mathcal{M}} I_m \cdot TC^-_{tm}\\ &+\alpha_2 \cdot \left(TC^+_{min} - TC^-_{min}\right)\\ &+\alpha_3 \cdot WO\end{aligned} \tag{5}$$

with weights $\alpha_1 \gg \alpha_2 \gg \alpha_3$. While we solve the MILP model using lexicographic optimization in this paper, the weighted objective function described above is used as part of fitness functions within different phases of the proposed heuristic and is included here for completeness.

3.5. Constraints

The objective function is subject to the following constraints.

3.5.1. Flight initiation constraints

$$\sum_{t\in\mathcal{T}}\sum_{m\in\mathcal{M}} c^t_{km} \leq N_k,\ \forall k \tag{6}$$

$$\sum_{m\in\mathcal{M}} c^t_{km} \leq 1,\ \forall k,t \tag{7}$$

$$\begin{aligned}&\sum_{t'=0}^{T_k+R_k-1}\ \sum_{m\in\mathcal{M}} c^{t+t'}_{km} \leq 1,\\ &\qquad \forall k, t \in \left\{1,\dots,T-(T_k+R_k-1)\right\}\end{aligned} \tag{8}$$

$$c^t_{km} = 0,\ \forall k,m,t \in \left\{T-(T_k-2),\dots,T\right\} \tag{9}$$

Constraint (6) ensures that the number of flights initiated by resource k does not exceed the maximal number of flights permitted for that resource. Constraint (7) allows a resource to only initiate a flight towards one area of operation in a given time slot. Since we assume that resources have a fixed flying and minimal resting time, constraint (8) ensures that a resource can initiate at most one flight in a time window corresponding to the flight time and rest time. Constraint (9) ensures that a resource does not initiate a flight in posterior time slots when it cannot complete the full flight.

3.5.2. Phases of flight constraints

$$e^{t+U_{km}}_{km} \geq c^t_{km},\ \forall k,m,t \in \left\{1,\dots,T-(T_k-1)\right\} \tag{10}$$

$$e^{t+(T_k-1)-U_{km}}_{km} \geq c^t_{km},\ \forall k,m,t \in \{1,\dots,T-(T_k-1)\} \tag{11}$$

$$\begin{aligned}&x^{t+t'}_{km} \geq c^t_{km},\ \forall k,m,t \in \left\{1,\dots,T-(T_k-1)\right\},\\ &\qquad t' \in \left\{U_{km}+1,\dots,T_k-U_{km}-2\right\}\end{aligned} \tag{12}$$

$$2 \cdot \sum_{t\in\mathcal{T}} c^t_{km} = \sum_{t\in\mathcal{T}} e^t_{km},\ \forall k,m \tag{13}$$

$$\left(T_k - 2\cdot U_{km} - 2\right)\cdot \sum_{t\in\mathcal{T}} c^t_{km} = \sum_{t\in\mathcal{T}} x^t_{km},\ \forall k,m \tag{14}$$

$$x^t_{km} + e^t_{km} \leq 1,\ \forall k,m,t \tag{15}$$

Constraints (10)–(15) are used to determine the different phases of flight, i.e., whether a resource is arriving/leaving an area or fighting the fire in a given time slot according to the flight initiation schedule. Constraints (10), (11), and (12) ensure resources are arriving, leaving and fighting the fire in their assigned area of operation, respectively, according to their transit and take-off times. Constraints (13) and (14) ensure that arriving/leaving and firefighting time slots, respectively, can take value 1 only if the flight is initiated, and constraint (15) ensures resources cannot be arriving/leaving and fully firefighting in the same time slot.

3.5.3. Resource activity constraints

$$s_{km}^{t+t'} \geq c_{km}^{t},\ \forall k, m, t \in \{1, \ldots, T - (T_k - 1)\},\quad t' \in \{0, \ldots, T_k - 1\} \tag{16}$$

$$T_k \cdot \sum_{t\in\mathcal{T}} c_{km}^{t} = \sum_{t\in\mathcal{T}} s_{km}^{t},\ \forall k, m \tag{17}$$

$$\sum_{m\in\mathcal{M}} s_{km}^{t} \leq A_k^t,\ \forall k, t \tag{18}$$

$$M \cdot \left(1 - \sum_{m\in\mathcal{M}} s_{km}^{t}\right) \geq \sum_{t'=P_k}^{T-t} \sum_{m\in\mathcal{M}} s_{km}^{t+t'},\quad \forall k, t \in \{1, \ldots, T - P_k\} \tag{19}$$

Constraints (16) and (17) are used to set the time slots in which each resource is in a flight associated with a specific area of operation. The flight covers time slots needed for two-way transit, arriving/leaving the area and fighting the fire and is determined by the fixed flight time. Constraint (18) ensures that a resource can only be active if it is available in that time slot, while constraint (19) ensures that all flights are within a time window determined by the limit on the physical presence of the crew.

3.5.4. Area of operation constraints

$$\sum_{k\in\mathcal{K}} \left(x_{km}^{t} + e_{km}^{t}\right) \leq S_m,\ \forall m, t \tag{20}$$

$$v_{qm}^{t} \geq V_{qk} \cdot \left(x_{km}^{t} + e_{km}^{t}\right),\ \forall m, t, q, k \tag{21}$$

$$\sum_{q\in\mathcal{Q}} v_{qm}^{t} \leq 1,\ \forall m, t \tag{22}$$

$$v_{qm}^{t} \geq B_{qm},\ \forall m, q, t \tag{23}$$

Constraint (20) ensures that the number of resources flying at the same area of operation in the same time slot does not exceed the allowed limit. Constraints (21) and (22) ensure that only resources of the same type are in a single area at the same time. Constraint (23) ensures that resources are assigned to areas in which they can operate according to type.

3.5.5. Target completion and water output constraints

$$TC_{tm}^{+} - TC_{tm}^{-} = \sum_{k\in\mathcal{K}} C_k \cdot E_{km}^{t} \cdot e_{km}^{t} + \sum_{k\in\mathcal{K}} C_k \cdot D_{km}^{t} \cdot x_{km}^{t} - W_m^t,\ \forall t, m \tag{24}$$

$$TC_{min}^{+} - TC_{min}^{-} \leq TC_{tm}^{+} - TC_{tm}^{-},\ \forall t, m \tag{25}$$

$$WO = \sum_{k\in\mathcal{K}} \sum_{m\in\mathcal{M}} \sum_{t\in\mathcal{T}} C_k \cdot \left(E_{km}^{t} \cdot e_{km}^{t} + D_{km}^{t} \cdot x_{km}^{t}\right) \tag{26}$$

Constraint (24) calculates the water target completion (surplus or deficit) in each time slot at each area of operation. Constraint (25) calculates the minimal water target completion over all time slots and areas of operation and, finally, constraint (26) calculates the total water output. Note, the number of downloads of a resource depends on the area and time slot in which it is active. Consequently, different schedules with the same number of resources performing the same number/duration of flights can result in different total water outputs.

3.6. Formulation size

Herein, we discuss the overall size of the formulation in terms of the number of variables and constraints. To simplify notation, aliases for the sizes of sets are defined: $T = |\mathcal{T}|$, $K = |\mathcal{K}|$, $Q = |\mathcal{Q}|$, and $M = |\mathcal{M}|$. The total number of variables is given by: $4TKM + TQM + 2TM + 2$, dominated by the first element $4TKM$. If average values for T_k, R_k, P_k, and U_{mk} are denoted as $\overline{T_k}$, $\overline{R_k}$, $\overline{P_k}$, $\overline{U}$, respectively, the total number of constraints is: $K(4T - \overline{T_k} - \overline{R_k} - \overline{P_k} + 2) + KMT(2\overline{T_k} + Q - 2\overline{U} + 1) - KM\overline{T_k}(2\overline{T_k} + 2\overline{U} - 3) - 2KM(\overline{U} - 1) + MT(Q + 4) + 1$. The number of constraints is highly dominated by the term $2KMT\overline{T_k}$, which provides a good approximation of the problem size.

4. Heuristic approach: It-RG-SA

While the MILP model described in the previous section can find optimal solutions for smaller instances, it is intractable for larger cases. Herein we describe an efficient heuristic approach based on an iterative multi-start approach combining a randomized greedy algorithm with simulated annealing. The proposed approach is a modified version of Greedy Randomized Adaptive Search Procedure (GRASP), a multi-start metaheuristic algorithm first proposed in [45]. In GRASP, each iteration (or independent start) consists of two phases: (i) initial construction based on a randomized greedy algorithm, and (2) local search applied to find the local optimum departing from the constructed solution [46]. In our approach, the construction phase applies an RG algorithm as in GRASP, but the local search phase is replaced with simulated annealing employing large neighborhood search techniques for neighbor generation. SA allows the search to escape local optima while LNS techniques enable efficient navigation between feasible solutions even though the problem is highly constrained. The pseudocode of the proposed algorithm, referred to as It-RG-SA, is given in Algorithm 1.

A valid solution or schedule is given as a set of takeoffs and denoted as S. Takeoffs are defined as tuples $c = (k, m, t)$, corresponding to decision variables c_{km}^{t}. If a set of takeoffs does not satisfy the constraints, it is not considered to be a valid schedule. Note, every subset of a valid schedule is also a valid schedule (including an empty set), but is always an inferior solution. We limit ourselves to operating exclusively on valid schedules throughout the heuristic algorithm by permitting the removal of any subset of takeoffs, while allowing the addition of only feasible takeoffs. The algorithm performs multiple independent iterations, where each iteration runs the RG and SA phases consecutively, and the best found solution, S^{best}, is returned as the final schedule. Descriptions of the individual RG and SA phases of It-RG-SA are given in the following subsections.

4.1. The RG phase

The pseudocode of the proposed RG algorithm used in It-RG-SA is given in Algorithm 2. The algorithm starts from an initial solution S and gradually inserts takeoffs into the schedule until no more takeoffs can be inserted. In the construction phase of the It-RG-SA algorithm, the initial solution is an empty set. However, this algorithm is also used to add takeoffs to existing solutions for neighborhood generation within SA, so an initial solution is given as an input parameter.

To build a valid schedule, a candidate set of feasible takeoffs is created, C, comprised of all potential takeoffs currently not included in the solution ($C \cap S = \emptyset$), whose addition to the schedule would not breach the constraints (line 2). All candidate elements, i.e., feasible takeoffs, are then evaluated based on a fitness function, $g(c)$. Note, this function is not the same as the objective function, but rather is a weighted combination of the weighted objective function ($f(c)$) and two additional metrics ($p(c)$ and $r(c)$) designed to avoid choosing takeoffs in the current iteration which could severely limit the number of choices in subsequent iterations. This to some extent softens the myopic aspect of a purely objective-based greedy approach. The first

Algorithm 1 Iterative Randomized Greedy and Simulated Annealing (It-RG-SA)

1: **procedure** IT-RG-SA($Iter, N_D, \alpha_G, \gamma, \alpha_D, T_0, \beta, L$)
2: $S^{\text{best}} \leftarrow \emptyset$;
3: **for** $i \leftarrow 1$ **to** $Iter$ **do**
4: $S \leftarrow \emptyset$;
5: $S' \leftarrow$ RANDOMIZED_GREEDY(S, α_G, γ);
6: $S'' \leftarrow$ SIMULATED_ANNEALING($S', N_D, \gamma, \alpha_D, \alpha_G, T_0, \beta, L$);
7: **if** $f(S'') > f(S^{\text{best}})$ **then**
8: $S^{\text{best}} \leftarrow S''$;
9: **end if**
10: **end for**
11: **return** S^{best};
12: **end procedure**

Algorithm 2 Randomized Greedy (RG)

1: **procedure** RANDOMIZED_GREEDY(S, α_G, γ)
2: $C \leftarrow$ Candidate set of all feasible takeoffs which could be added to S without breaching constraints;
3: **while** $C \neq \emptyset$ **do**
4: **for** all $c \in C$ **do**
5: $f(c) \leftarrow$ Weighted objective function $\mathcal{F}_w$ value for schedule $S \cup \{c\}$;
6: $p(c) \leftarrow$ Number of feasible takeoffs (size of updated set C) for schedule $S \cup \{c\}$;
7: $r(c) \leftarrow$ Number of takeoffs in C which involve the same resource as c, $\left|\left\{c' \in C : k(c') = k(c)\right\}\right|$;
8: **end for**
9: $\hat{f}(c)$, $\hat{p}(c)$, $\hat{r}(c) \leftarrow f(c)$, $p(c)$, $r(c)$ linearly scaled to range $[0, 1]$;
10: $w = \gamma \cdot \sqrt{F - |S| - 1}$, where $F = \sum_{k \in \mathcal{K}} N_k$;
11: **for** all $c \in C$ **do**
12: $g(c) = \hat{f}(c) + w \cdot (\hat{p}(c) - \hat{r}(c))$;
13: **end for**
14: $g^{min} \leftarrow \min\{g(c) \mid c \in C\}$;
15: $g^{max} \leftarrow \max\{g(c) \mid c \in C\}$;
16: $RCL \leftarrow \left\{c \in C \mid g(c) \geq g^{max} - \alpha_G \left(g^{max} - g^{min}\right)\right\}$;
17: Randomly select c' from RCL;
18: $S \leftarrow S \cup \{c'\}$;
19: Update C for S with respect to constraints;
20: **end while**
21: **return** S;
22: **end procedure**

Algorithm 3 Simulated Annealing (SA)

1: **procedure** SIMULATED_ANNEALING($S, N_D, \gamma, \alpha_D, \alpha_G, T_0, \beta, L$)
2: $S^{\text{best}} \leftarrow S$;
3: $T \leftarrow T_0$;
4: **for** $i \leftarrow 1$ **to** L **do**
5: Destroy: $S_D \leftarrow$ DESTROY(S, N_D, α_D);
6: Repair: $S_R \leftarrow$ RANDOMIZED_GREEDY(S_D, α_G, γ);
7: **if** $f(S_R) > f(S^{\text{best}})$ **then**
8: $S^{\text{best}} \leftarrow S_R$;
9: **end if**
10: $\Delta f = f(S_R) - f(S)$;
11: **if** $\Delta f > 0$ **or** $\text{rand}(0,1) < e^{\Delta f / T}$ **then**
12: $S \leftarrow S_R$;
13: **end if**
14: $T \leftarrow \beta \cdot T$;
15: **end for**
16: **return** S^{best};
17: **end procedure**

metric, $f(c)$, calculates the value of the weighted objective function $\mathcal{F}_w$ (given in (5)) for the tentative solution if takeoff c were inserted into the current schedule (line 5). The second metric, $p(c)$, gives the total number of remaining feasible takeoffs (updated set C) if takeoff c were added to the current solution S (line 6). Higher values are considered better because this indicates that adding the takeoff does not limit the number of options in subsequent steps as much as those takeoffs with lower values. The third metric, $r(c)$, counts the number of

Algorithm 4 Destroy Method

1: **procedure** DESTROY(S, N_D, α_D)
2: $N'_D \leftarrow$ Random integer from $\{1, 2, \ldots, \min(N_D, |S|)\}$;
3: **for** $i \leftarrow 1$ **to** N'_D **do**
4: **for** all $c \in S$ **do**
5: $q(c) \leftarrow$ Weighted objective function $\mathcal{F}_w$ value for $S \setminus \{c\}$;
6: **end for**
7: $q^{min} \leftarrow \min\{q(c) \mid c \in S\}$;
8: $q^{max} \leftarrow \max\{q(c) \mid c \in S\}$;
9: $RCL \leftarrow \left\{c \in S \mid q(c) \geq q^{max} - \alpha_D \left(q^{max} - q^{min}\right)\right\}$;
10: Randomly select c' from RCL;
11: $S \leftarrow S \setminus \{c'\}$;
12: **end for**
13: **return** S;
14: **end procedure**

distinct potential takeoffs $c' \in C$ which contain the same aerial resource k as takeoff c (line 7), where the resource of a takeoff c is denoted as $k(c)$. Takeoffs with lower values are preferred because the associated resource has fewer takeoffs to choose from. Consequently, it is more likely that, if not chosen in the current iteration, a resource may be left without feasible takeoffs in later stages. All three metrics are linearly scaled to the range $[0, 1]$ (line 9) and combined in fitness function $g(c)$ according to weight w (line 12). Weight w is tuned by parameter γ and dynamically adjusted such that the influence of metrics r and p is high for near-empty solutions, but is gradually reduced as solution construction advances, such that the objective function plays a more significant role when only a few feasible takeoffs remain (line 10).

Based on the fitness function, a Restricted Candidate List (RCL) is formed containing all feasible takeoffs whose fitness is greater than or equal to a threshold. The threshold is based on the fitness of all feasible takeoffs and a parameter $\alpha_G \in [0, 1]$ (lines 14–16). If α_G is set to 0, only the best takeoff(s) enter the RCL (pure greedy), while for a value of 1 all feasible takeoffs are included. A takeoff is then randomly selected from the RCL (line 17), added to the schedule (line 18) and the candidate list of feasible takeoffs is updated (line 19). The algorithm stops when there are no more feasible takeoffs available (line 3) and schedule S is returned as the final solution (line 21).

4.2. The SA phase

To improve upon the solution constructed by the RG algorithm, SA is applied. SA is an iterative, single-solution based metaheuristic founded on ideas from statistical mechanics and a Monte Carlo model which simulates energy levels in cooling solids [47]. The algorithm seeks to improve a current solution by sampling its neighborhood, i.e., the set of solutions generated from the current one by applying small local perturbations. SA is a probabilistic technique which evaluates a single neighbor in each iteration. In contrast to hill-climbing, non-improving solutions in SA have a non-zero probability of acceptance allowing the search to escape from local optima. The probability of acceptance of non-improving neighbors depends on their relative fitness compared to the current solution and a control parameter, T, representing temperature in the real-world annealing process. Parameter T is initially high, allowing for a higher probability of accepting non-improving solutions to support diversification of the search. To facilitate convergence, the temperature is reduced as the search advances, intensifying the search in a local area. If an evaluated neighbor improves the current solution, it is typically selected with a probability of 1. The pseudocode of the Simulated Annealing algorithm used is given in Algorithm 3. A homogeneous cooling schedule is applied, slightly decreasing the temperature in every iteration, starting from an initial temperature T_0 and using a geometric decrement function: $T \leftarrow \beta \cdot T$. Cooling factor $\beta \in [0, 1]$ determines the cooling speed and the algorithm is run for L iterations.

In the proposed SA algorithm, in order to generate the neighboring solution in each iteration, a single neighbor is sampled from a large neighborhood defined using LNS techniques. LNS, initially proposed in [48], is a metaheuristic algorithm where, instead of defining an explicit neighborhood, the neighborhood is defined implicitly by a pair of *destroy* and *repair* methods. The *destroy* method destroys part of the current solution, typically containing an element of stochasticity, resulting in a partial solution which may or may not be feasible. The partial solution is passed to the *repair* method to be made feasible and/or improved, but usually different from the starting solution. The neighborhood of a solution is defined as the set of solutions that can be reached by applying the destroy and repair methods. This type of large neighborhood allows the search to navigate among feasible solutions even if the problem is highly constrained.

In our SA algorithm, we employ a *destroy* procedure which removes a subset of takeoffs from the current schedule (line 5). Recall, removing takeoffs from a valid schedule always produces a valid, but inferior, solution. Since adding any subset of takeoffs which do not breach the constraints will always improve upon a reduced schedule, we subsequently *repair* or, more precisely, *improve* the reduced schedule by adding takeoffs until no further feasible takeoffs exist. This is done by utilizing the randomized greedy algorithm described in the previous subsection (line 6).

The pseudocode of the *destroy* method is shown in Algorithm 4. The solution is partially destroyed by removing up to N_D (or $|S|$) takeoffs. Takeoffs in the current schedule, $c \in S$, are ranked in inverse order according to their contribution to the solution quality. The fitness function, $q(c)$, is calculated based on the value of the weighted objective function of the reduced solution obtained by removing takeoff c (line 5). Takeoffs with lower calculated fitness values contribute more to the objective function value in the current (non-reduced) solution, indicating that it may not be desirable to remove them. An RCL is comprised of the takeoffs which contribute least to the current objective function, i.e., have the highest $q(c)$ values (line 9). One is then selected at random to be removed from the solution (lines 10–11). The size of the RCL is controlled via parameter α_D, where a value of 1 means a takeoff is chosen completely at random while a value of 0 means the "worst" takeoff is always removed. After removing a takeoff, the solution is updated and remaining takeoffs are re-evaluated. The process continues until N'_D takeoffs are removed.

5. Computational results

In this section, we perform a computational study to evaluate the efficiency of the proposed approaches.[2] Details of the test scenarios and

[2] The full results, containing all input and output files, can be found in the following public *GitHub* repository: https://github.com/LMesaric/Aerial-Resource-Scheduler.

Table 2
Resource characteristics and parametrization of test scenarios.

Aircraft types ($\mathcal{K}$)	Sub-type	Capacity (l)	Average drops/h	Models	TS1	TS2	TS3	TS4	TS5	TS6	TS7	TS8
Helicopter (rotary-wing)	Light	900	5	AS350B3, A119 Koala	2	3	5	7	8	10	11	17
	Semi-1	1500	5	Bell 412, W-3 Sokól	1	1	2	3	3	4	4	6
	Semi-2	2100	5	AS332 Super Puma, NH90	0	1	0	0	2	1	3	4
	Heavy	4500	4	Kamov Ka-32	1	1	2	3	3	4	4	6
Airplane (fixed-wing)	Amphibious	5500	3	CL-215, CL-415	3	4	6	7	9	11	13	17
Total resources ($\lvert\mathcal{K}\rvert$)					7	10	15	20	25	30	35	50
Areas ($\lvert\mathcal{M}\rvert$)					2	3	3	4	4	5	5	8

Table 3
Regulation restrictions.

Resource type (Q)	T_k	R_k	N_k	P_k
Helicopter (rotary-wing)	6	2	4	36
Airplane (fixed-wing)	12	4	2	36

input data are given in Section 5.1, while implementation details, the simulation setup and parameter tuning are described in Section 5.2. Section 5.3 compares results of the proposed heuristic approach to the MILP model, evaluates individual phases and iterations of the heuristic and discusses scalability issues.

5.1. Test data

To generate realistic test data, we focused on current regulations, historical data, and resources deployed in Spain. To test algorithm robustness and scalability for different fire situations, a diverse set of test data was generated with varying sizes and water target distributions according to distinct extinction strategies. These strategies are designed to simulate different topographical and meteorological conditions, varying degrees of potential damage and/or casualties in specific areas, and subjective decisions made by the fire manager.

5.1.1. Problem sizes, resources and regulations

We assume daylight hours are between 7:00 and 22:00, which is representative of a typical summer day in Spain. This results in 15 h of daytime, split into 45 time slots, each 20 min long: $\mathcal{T} = \{1, \ldots, 45\}$. The time slot duration was set to 20 min since (i) most regulatory restrictions are in multiples of 20 minutes [44] and (ii) according to historical data, most resources stationed in the same province can arrive to a fire location in the province within 20 minutes [9]. Resources are divided into two types (set $\mathcal{Q}$): helicopters (rotary-wing) and airplanes (fixed-wing), which should not be assigned simultaneously to the same area of operation in accordance with [49]. Different sub-types are also considered as outlined in Table 2 with varying capacities and models from [50]. Flight duration (T_k), minimal duration of resting periods between continuous flights (R_k), maximal number of flights per day (N_k), and maximal crew physical presence time (P_k) were selected according to Spanish regulations for civil fire-fighting aviation [44] divided per type and outlined in Table 3.

To simulate mega-fires of different intensities, 8 different test scenarios (TS) were considered as shown in Table 2, ranging from smaller mega-fires mobilizing seven aerial resources over two areas of operation, to huge fires with 50 resources allocated to eight different firefighting areas. For each test scenario, eight different water target distribution strategies were considered, with five instances each, as will be described in Section 5.1.3. In four of the problem instances for each water target distribution strategy, all areas of operation were considered equally important and assigned an importance parameter value of $I_m = 1$. In the 5th instance, one or two areas were selected randomly as high priority areas and assigned values $I_m = 2$, while the remaining areas were assigned $I_m = 1$.

The distribution of types (and helicopter sub-types) of aerial resources for different problem sizes was selected to be roughly in line with Spain's Forest Fire Campaign for 2022, focused on the autonomous community of Andalusia where the largest number of large-scale forest fires occurs [51,52]. Helicopters make up around 65% of all aerial resources in each test scenario, while airplanes account for the remaining 35%. Helicopters are further divided by subtypes light, semi-1 (civilian), semi-2 (military), and heavy categories, with approximately 50%, 20%, 20%, and 10% shares, respectively. We only consider amphibious airplanes, but fixed-wing aircraft which load on the ground, such as Air Tractors AT-802, could be added as a separate type or a sub-type of airplane, depending on the restrictions related to sharing flight paths.

5.1.2. Area limitations, drops, transit times, availability

The maximum number of aircraft that can be assigned to a single area of operation to fly in carousel (S_m), was chosen at random in the range $[7, 10]$ for each fire area and each test instance, as values may vary based on terrain features, such as the number of water points and the size of the area. Helicopters are assumed to be able to operate in all areas, but some areas may not be accessible to airplanes. Each test instance was generated with a 20% chance of containing one such area (selected at random) where only helicopters are allowed to operate (parameter B_{qm}).

The average number of drops per time slot for each resource in each area and time slot (D^t_{km}) was derived from estimated values on the average number of drops per hour for different sub-types as given in Table 2. The average drops per hour for helicopters was selected according to data from [53,54], while the average number of drops for airtankers was selected based on historical data provided by the 43rd Grupo (Group) Firefighting Squadron of the Spanish Air Force for years 2020 and 2021. The actual number of drops in each area and each time slot will depend on factors such as distance to water points in each area and meteorological conditions, such as increased wind speed, over the course of the day [54]. Consequently, to set values D^t_{km} in each test instance, the average number of drops from Table 2, normalized per time slot, is modified up to $\pm 20\%$ for each area according to a random accessibility rating assigned to each area. To simulate varying weather conditions over time, the number of drops between 15:00 and 18:00 is additionally reduced by 5%.

According to historical data for the Andalusia region, most resources stationed within the same province can reach the fire within 20 min, while resources stationed in other provinces tend to have transit times ranging from 15 min to a maximum of 1 h 10 min [9]. Since helicopters are more flexible with respect to takeoff and landing, they can be stationed at a larger set of destinations (airports, bases, fields, or even parking lots). Furthermore, they have more limited autonomy and flight duration, making longer transit times unreasonable. Consequently, helicopters are assumed to be able to reach any area of the fire within a single 20 min time slot making the number of pure transit slots (U_{km}) zero. The transit time is then reflected in parameter E^t_{km}, indicating the average number of drops the resource can make when it is arriving/leaving the fire. These values were selected to be between 0% and 50% of their firefighting counterpart (D^t_{km}), with up to 5% variation between areas of the fire, to simulate the fact that part of the time is spent on transit, as opposed to exclusively fighting the fire.

As opposed to helicopters, airplanes require a runway, making the set of possible bases more limited, and have longer maximum flight

Table 4
Water target distribution variations.

Correction factor (CF)	Distribution over areas	Distribution over time
Low (25%)	Uniform (UA)	Initial Attack (IA)
High (50%)	Non-uniform (NA)	Mostly-uniform (MUT)

duration, making longer transit times less of an issue. Thus, airplanes are assumed to arrive at the fire within 1 h. The number of pure transit time slots U_{km} was set randomly to 0, 1, or 2 for each airplane. The remaining transit time is reflected in parameter E^t_{km} as for helicopters. Note, individual areas are assumed to be relatively close to each other, thus, the same value of U_{km} is set for all areas for a given resource, while the difference in transit to individual areas is reflected in parameter E^t_{km}.

With respect to availability, A^t_k, helicopters are assumed available in all time slots, as they typically remain in proximity to the fire and usually do not come from faraway bases. For airplanes, it is considered that at the start of the day they may have to fly in from a further base, or return to it at the end of the day, or perhaps to fight another wildfire the following day. Consequently, to generate the test instances, each fixed-wing resource has an 85% probability of being available all day, a 10% probability of being unavailable at the beginning of the day (up to a time slot within 10:20 – 13:00), and a 5% probability of being unavailable at the end of the day (starting between 18:20 – 20:00). If an airplane is unavailable for a part of the day, one flight is deducted from the available daily quota (N_k) as it is assumed to be transiting or fighting another fire in that period.

5.1.3. Water target distributions

To test the robustness of the algorithm under different conditions, we create diverse water target distributions (W^t_m) simulating different fire characteristics, priorities and extinction strategies. To do so, the estimated total water output (EWO) is first calculated by multiplying the maximum number of fully firefighting time slots for each resource with the average amount of water dropped in firefighting time slots, summed over all resources. This value is then (1) adjusted by a tightness correction factor, (2) distributed over fire areas, and (3) distributed over time slots. Each step considers two different possibilities, as outlined in Table 4, for a total of 8 different water target distribution schemes. For each test scenario, 5 instances are generated for each water target distribution scheme, giving 40 instances per test scenario and a total of 320 problem instances.

While the total water output WO will approach EWO, it cannot be distributed over area and time periods arbitrarily due to regulatory restrictions, granularity of resources, capacities, area characteristics, etc. Thus, to set accessible water targets in individual time slots and areas from the estimated total output, it must be adjusted accordingly. A tightness correction factor (CF) is applied to simulate different levels of estimated minimal water target per time slot and area. Note, estimating the actual required water target is not an easy task, and over/under estimation of the needed water and allocated resources may occur due to the uncertainty of conditions. A lower tightness correction factor sets a lower, more accessible, water target in individual time slots and areas which can be covered by the available resources to ensure a minimal continuity over time and space. The optimization then focuses on maximizing the TC_{min} and WO. A higher tightness correction factor sets a tighter water target in individual time slots and areas, where resources are assumed to be more scarce and may be under-allocated. We consider two tightness correction factors: Low (25%) and High (50%) indicating the percentage of EWO which will be distributed over areas and time to set water targets.

Once the estimated total water output (EWO) is adjusted by the tightness correction factor (CF), it is distributed either Uniformly over Areas (UA) or Non-uniformly over Areas (NA). Non-uniform distributions are meant to simulate prioritization of areas due to different fire intensities, topographical conditions, and/or varying degrees of potential damage and/or casualties in specific areas. For non-uniform distributions, the following proportions were used according to the number of areas: $|\mathcal{M}| = 2$: {65%, 35%}, $|\mathcal{M}| = 3$: {50%, 25%, 25%}, $|\mathcal{M}| = 4$: {40%, 25%, 20%, 15%}, $|\mathcal{M}| = 5$: {35%, 20%, 20%, 15%, 10%}, and $|\mathcal{M}| = 8$: {20%, 20%, 15%, 15%, 10%, 10%, 5%, 5%}.

After the water target values are distributed over areas, they must be distributed over time, giving the final W^t_m values. Two strategies are considered for time distribution: Initial Attack (IA) and Mostly Uniform over Time (MUT). IA will uniformly distribute 60% of the water target in the first six hours of the day, and the remaining water target across the remaining time slots, simulating a strong attack at the start of the day, but still maintaining a minimal continuity over time for the remainder of the day. MUT, on the other hand, distributes the water target uniformly over all time slots with 75% probability. For the remaining 25% probability, MUT will distribute the water target uniformly over all time slots except those between 13:00 and 18:20 which are assigned an approximately 15% higher target to simulate varying meteorological conditions, such as stronger winds which can increase the risk of the fire spreading.

5.2. Implementation and parameter tuning

The MILP formulation was modeled in AMPL and solved with CPLEX 22.1.1 using lexicographic optimization. The proposed heuristic was implemented in C++20 and compiled with GCC 12.2. Both approaches were executed on Google Compute Engine T2D series virtual machines with 4 vCPUs and 16 GB of memory, with disabled simultaneous multithreading (SMT), running Debian GNU/Linux 12.

The weights used by the heuristic for the weighted objective function $\mathcal{F}_w$ given in Eq. (5) were set to $\alpha_1 = 10^7$, $\alpha_2 = 10^2$, and $\alpha_1 = 10^{-5}$ which ensures that even 1 l (0.001 kl) of difference in one objective carries more weight than the biggest possible change in the next objective for the test data generated. To configure the remaining algorithm parameters for final experimentation, sequential parameter tuning was performed for individual phases of the It-RG-SA heuristic and the main results are summarized here. For each configuration, all 320 test instances were run and the average results for each objective are shown, along with the number of instances for which that parameter value reached the best primary objective value (including ties). Additionally, execution time is shown for parameters which significantly impact run time.

Recall, the problem maximizes three hierarchical objectives given in Eqs. (1)–(3). In the results, we show the individual objectives as follows: the primary objective is denoted as TD (Total Deficit) representing the sum of water deficit, weighted by the importance of each area, over all areas and time slots and given as a negative value, i.e., $TD = -\sum_{t\in\mathcal{T}}\sum_{m\in\mathcal{M}} I_m \cdot TC^-_{tm}$. A value of 0 indicates that the water target is achieved in all areas and time slots. The secondary objective representing the minimal target completion is denoted as TC_{min} (which can be positive or negative), and finally, the positive total water output is denoted as WO. All values are maximized and shown in kl for a sense of scale. Namely, the water capacities of the aerial resources range from 900 l to 5,500 l. Consequently, a difference of 1 kl in the value of an objective corresponds to approximately 1 light helicopter drop, while differences in the order of 5 kl correspond to 5–6 light helicopter drops or 1 heavy helicopter or amphibious airplane drop.

Sequential parameter tuning was performed as follows. To set multiplier γ in weight w in the Randomized Greedy (RG) phase of the algorithm, the 320 instances were run for 100 iterations of RG for varying γ values, as shown in Table 5. In these experiments, α_G was set to 0.1 based on preliminary testing. Recall, weight w is meant to dynamically adjust the influence of metrics r and p such that their influence is high for empty solutions and gradually reduced as solution construction advances. For $\gamma = 0$, the influence of these metrics is eliminated and only the weighted objective function is considered in

Table 5
Comparison of results for parameter γ for the RG phase.

γ	TD (TC_{min}, WO) [kl]	#best
0	−52.18 (−3.86, 1,203)	5
0.2	**-21.35 (-2.25, 1490)**	**291**
0.5	−43.21 (−3.13, 1,483)	23
1	−62.84 (−3.68, 1,491)	1

Table 6
Comparison of results for parameter α_G for the RG phase.

α_G	TD (TC_{min}, WO) [kl]	#best
0 (Greedy)	−23.28 (−2.50, 1,539)	99
0.1	**-21.77 (-2.28, 1488)**	**205**
0.25	−34.37 (−3.18, 1,474)	15
0.5	−43.65 (−3.68, 1,459)	1
1 (Random)	−90.93 (−4.28, 1,433)	0

Table 7
Comparison of results for parameter N_D for the SA phase.

N_D	TD (TC_{min}, WO) [kl]	#best	Time [min]
2	−20.67 (−2.17, 1,532)	4	31.28
5	−14.46 (−1.55, 1,536)	16	37.99
10	−13.13 (−1.32, 1,539)	40	61.36
15	−11.12 (−1.17, 1,541)	73	107.98
20	−8.76 (−1.12, 1,540)	84	188.02
30	−10.33 (−1.05, 1,536)	111	497.97

Table 8
Comparison of results for parameter α_D for the SA phase.

α_D	TD (TC_{min}, WO) [kl]	#best
0 (Greedy)	−37.85 (−3.48, 1,499)	0
0.5	−13.10 (−1.51, 1,541)	58
0.75	−11.83 (−1.18, 1,538)	80
0.9	**-9.99 (-1.07, 1539)**	**96**
1 (Random)	−10.23 (−1.07, 1,542)	92

Table 9
Comparison of results for parameters T_0 and β for the SA phase.

T_0	β	TD (TC_{min}, WO) [kl]	#best
10^{11}	0.990	−11.53 (−1.36, 1,540)	50
10^{11}	0.995	−11.74 (−1.29, 1,543)	46
10^{12}	0.990	−11.92 (−1.39, 1,540)	48
$\mathbf{10^{12}}$	**0.995**	**-10.27 (-1.17, 1538)**	**67**
10^{13}	0.990	−11.64 (−1.18, 1,541)	57
10^{13}	0.995	−10.34 (−1.10, 1,540)	62

Table 10
Configuration of It-RG-SA parameters used in final experiments.

γ	α_G	T_0	β	N_D	α_D	L
0.2	0.1	10^{12}	0.995	15	0.9	10^4

the greedy step. As γ is increased, the influence of these parameters is more significant. We can see that the best results were obtained for $\gamma = 0.2$ (in bold), finding the best solution in 291 out of the 320 problem instances. Consequently, this value was used in the remaining experiments.

The second experiment used to tune the parameters of the RG phase of the algorithm varied values for parameter α_G, from pure greedy to pure random. Results were obtained by re-running the 320 instances for 100 iterations of RG with $\gamma = 0.2$ and varying α_G values as shown in Table 6. We can see that a value of $\alpha_G = 0.1$ gave the best results, finding the best solution among all configurations in 205 out of the 320 cases.

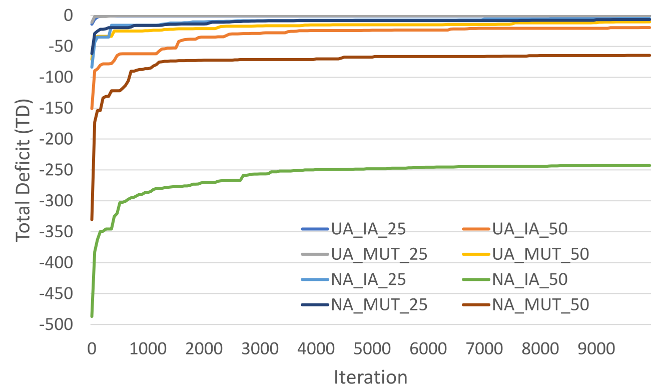

Fig. 1. Evolution of the Total Deficit over individual SA iterations for large instances (TS8).

To tune the parameters for the simulated annealing phase, SA was run starting from a greedy solution obtained with RG ($\alpha_G = 0.1$) with varying values for parameter N_D. The remaining parameters were determined according to several random configuration tests. Results for $L = 10^4$, $T = 10^{12}$, $\beta = 0.995$, and $\alpha_D = 0.9$ are shown in Table 7. Results for different configurations follow a similar trend. The total execution time needed to run all 320 instances is also shown. We can see that the results are consistently superior as N_D increases, i.e., for larger neighborhoods. However, this comes at a trade-off with significantly increased execution time. In order to run SA within the It-RG-SA algorithm and maintain a reasonable run time for the bigger instances and their application scope, an intermediate value of $N_D = 15$ was selected for further experiments. Note, run times vary significantly between instance sizes, e.g., from 4 s for TS1 to 50 s for TS8 with $N_D = 15$. Consequently, in practice, this parameter could be adjusted according to available time and problem size. Run times of individual phases of It-RG-SA and for different instance sizes will be discussed in the next subsection.

Further tests were run for SA, starting from an RG ($\alpha_G = 0.1$) solution, by varying parameter α_D with results shown in Table 8. The remaining parameters were set to $L = 10^4$, $T = 10^{12}$, $\beta = 0.995$, and $N_D = 15$. We can see that higher values of α_D give superior results, providing more diversity for neighbor creation in the destroy method. Consequently, value $\alpha_D = 0.9$ was selected for the final experiments. Different combinations of initial temperature T_0 and β were also tested with the aforementioned parameter settings and results are shown in Table 9. Note, high initial temperature and β values are needed as the absolute values of the objective function are large (in the order of $10^{10} - 10^{11}$) due to large weight α_1 in the objective function. Values $T_0 = 10^{12}$ and $\beta = 0.995$ were selected for the final experiments.

Additionally, the evolution of the solutions in individual iterations of SA was investigated. The evolution of the total deficit (TD) over the 10^4 SA iterations for the largest instances (TS8), one randomly selected for each water target distribution strategy, is shown in Fig. 1. We can see that improvement is very significant in the first 3k iterations. While solutions mostly converge after this, some additional improvements occur even after 8k or 9k iterations. Consequently, the number of iterations in the final experiments was set to $L = 10^4$ to cover such improvements. The final configurations used for the results in the next subsection are shown in Table 10. The number of total iterations for It-RG-SA is varied in the results to investigate algorithm evolution.

5.3. Results

Herein we compare the results of the MILP and the It-RG-SA heuristic, evaluate the improvement achieved by individual phases and iterations of It-RG-SA, and discuss execution time. The MILP was run with a cutoff time of 4 h per instance, measured as elapsed real time, and was able find optimal solutions for 62 of the smaller test cases: all 40 instances of TS1, 21 instances of TS2, and 1 instance of TS3. Results for these test instances are shown in Table 11, along with the number of variables and constraints for each instance.[3] It-RG-SA was run for 1, 20 and 150 iterations and cases for which the heuristic found optimal solutions are marked in bold. We can see that the It-RG-SA found the optimal solution for all three objectives in 8, 24, and 27 out of the 62 cases when run for 1, 20 and 150 iterations, respectively. Running more iterations of It-RG-SA consistently yields superior results, although improvements are more significant when increasing the number of iterations from 1 to 20, than 20 to 150.

The average and standard absolute deviations are also shown at the bottom of the table. Note, the deviation for WO is positive, as It-RG-SA tends to find solutions with higher WO at the expense of slightly inferior primary and secondary objectives. We can see that both average and standard deviation are reduced as a higher number of iterations are run, but are small in all cases. Specifically, average deviations from the optimal TD values are -0.68 kl, -0.31 kl, and -0.19 kl for 1, 20, and 150 iterations of It-RG-SA, respectively. Corresponding TC_{min} deviations are -0.20 kl, -0.12 kl, and -0.07 kl. In practical terms, these deviations are a fraction a single light helicopter drop and could be considered marginal considering total water outputs range from 368 kl to 988 kl.

To investigate results for all problem sizes, the 320 instances corresponding to test scenarios TS1-TS8 were solved by running the following:

(a) MILP: 4 h cutoff time
(b) Greedy: the RG algorithm ($\alpha_G = 0$)
(c) RG: the RG algorithm ($\alpha_G = 0.1$)
(d) It-RG(100): 100 iterations of RG
(e) It-RG-SA(1): 1 iteration of It-RG-SA
(f) It-RG-SA(20): 20 iterations of It-RG-SA
(g) It-RG-SA(150): 150 iterations of It-RG-SA

Although the MILP is timed-out for larger instances, it is run to find feasible sub-optimal solutions, possibly of high quality. The greedy approach is meant to evaluate the effectiveness of a pure deterministic constructive approach. RG and It-RG(100) are run to evaluate the RG phase of the algorithm, while a single iteration of It-RG-SA investigates the effectiveness of SA as a means to improve upon a solution obtained by RG. Running It-RG-SA for 20 and 150 iterations investigates the improvement of the solutions for an increased number of iterations which may be run for different application scopes, such as early-morning planning (more time-sensitive) or night-before planning (less time-sensitive).

Results are grouped according to problem size (TS), water target distribution and correction factor, averaged over 5 instances for each combination. While varying water target distributions investigates the robustness of the approach for different extinction strategies, CF variations help investigate solution quality under tighter and weaker target requirements. Since possible target completion depends highly on CF, results for $CF = 25\%$ and 50% are separated, shown in Tables 12 and 13, respectively, with the best solution marked in bold. Note, in some cases TC_{min} may be positive even if TD is negative since results are averaged over five instances.

For $CF = 25\%$, we can see from Table 12 that for smaller test scenarios (TS1-TS4), the MILP performed best, while for most larger test scenarios (TS5-TS8), It-RG-SA(150) found superior solutions. Among the heuristic approaches, as expected, It-RG-SA(150) performed best in all cases. The MILP was able to meet all water targets ($TD = 0$) for 105 out of the individual 160 instances while It-RG-SA was able to do so for 71, 97 and 108 instances when run for only 1, 20 and 150 iterations, respectively. RG, Greedy and RG(100) were able to meet all water targets for only 0, 4, and 8 instances, respectively.

While a single run of RG produces inferior solutions compared to a deterministic pure greedy approach, running 100 iterations of RG consistently gives better results supporting its effectiveness in the construction phase of the iterative approach It-RG-SA. Furthermore, we can see that running even a single iteration of It-RG-SA significantly improves upon RG, as well as It-RG(100), indicating the effectiveness of the SA phase. When comparing multiple iterations of It-RG-SA, we can see that the main improvement is achieved in most cases with 20 iterations, while running 150 iterations brings only marginal improvements. Results do not vary significantly between target distributions indicating the robustness of the approach for different extinction strategies.

For tighter water target requirements, i.e., $CF = 50\%$, similar trends can be observed in Table 13: (i) the MILP finds superior solutions for smaller instances up to TS3, while It-RG-SA(150) performs best for TS4-TS8; (ii) It-RG-SA(1) significantly improves upon It-RG(100) solutions demonstrating effectiveness of the SA phase; (iii) the main improvement of multiple iterations of It-RG-SA is achieved after only 20 iterations; and (iv) similar trends are observed for all water target distributions. As each variation was run separately, in the case of TS6 NA-MUT, 20 iterations of It-RG-SA gave a better result than 150 iterations, but in general solution quality improves with the number of iterations.

For $CF = 50\%$, the MILP was able to find solutions within 1 kl of total deficit (∼1 light helicopter drop) for 58 out of the 160 instances, while the It-RG-SA algorithm was able to do so for 32, 80 and 87 instances when run for 1, 20 and 150 iterations, respectively. Furthermore, the MILP was able to find solutions within 5 kl (∼1 heavy helicopter or airplane drop) for 117 instances, while the It-RG-SA algorithm was able to do so for 100, 130 and 134 instances when run for 1, 20 and 150 iterations, respectively. RG, Greedy and RG(100) were able to find solutions within 5 kl of total deficit for only 1, 5, and 23 instances, respectively. For TS8, obtained solutions have large total deficits for a non-uniform distribution over areas, but we can see significant improvement as additional iterations of It-RG-SA are run. Since the optimal solution is unknown, it is not clear if the larger deficits for these cases are due to more constrictive test instances or poorer algorithm results. The algorithm was run for 300 iterations for these instances and no significant improvement was achieved, indicating that these cases are presumably more constrictive.

For easier visualization of the results, we plot the values for TD and TC_{min} for different problem sizes (TS) and CF values, averaged over all distributions in Figs. 2 and 3, respectively. The largest test scenario is omitted from the graphs for $CF = 50\%$ for better visibility of the results, but trends are analogous (with significant deterioration of MILP results) and can be seen in Table 13. Results for inferior solutions produced by RG are also omitted to maintain a smaller range of values but can be seen in Tables 12 and 13. We can see that It-RG(100) outperforms the pure greedy approach for both TD and TC_{min}. Furthermore, we can see that the It-RG-SA algorithm consistently improves upon It-RG(100), demonstrating effectiveness of the SA phase, particularly for larger test instances and tighter water targets ($CF = 50\%$). We also see that multiple iterations of It-RG-SA significantly improve results compared to a single iteration for both objectives, while improvements beyond 20 iterations are marginal, except for the largest test cases. The MILP gives high quality results for smaller test scenarios, but degrades as problem size increases, particularly for $CF = 50\%$. This is even more pronounced for TS8 ($CF = 50\%$) which can be seen in Table 13.

Since total WO values vary significantly between test scenarios, Fig. 4 shows the relative total water output of the solutions obtained by the MILP, Greedy and It-RG-SA algorithms in comparison to the water output obtained with It-RG(100). We can see that the MILP

[3] The number of variables and constraints corresponds to the adjusted problem obtained with the CPLEX *presolve* function which looks for logical reduction opportunities to reduce the problem size.

Table 11
Results for problem instances for which the MILP found optimal solutions, given as TD (TC_{min}, WO) [kl].

Instance	Variables	Constr.	MILP	It-RG-SA(1)	It-RG-SA(20)	It-RG-SA(150)
TS1-NA-IA-25-I1	2,182	8,033	**−1.38 (−0.69, 543)**	−1.47 (−0.69, 544)	**−1.38 (−0.69, 543)**	**−1.38 (−0.69, 543)**
TS1-NA-IA-25-I2	2,719	9,972	**−1.62 (−0.53, 368)**	−3.48 (−0.65, 426)	−2.39 (−0.69, 428)	−1.63 (−0.69, 428)
TS1-NA-IA-25-I3	2,779	10,883	**−0.10 (−0.09, 524)**	−0.11 (−0.09, 524)	**−0.10 (−0.09, 524)**	**−0.10 (−0.09, 524)**
TS1-NA-IA-25-I4	2,779	11,019	**−0.14 (−0.14, 448)**	−0.47 (−0.46, 470)	−0.47 (−0.46, 470)	−0.39 (−0.14, 462)
TS1-NA-IA-25-I5	2,751	10,450	**−0.88 (−0.22, 304)**	−1.87 (−0.67, 388)	−1.44 (−0.37, 380)	−1.43 (−0.37, 379)
TS1-NA-IA-50-I1	2,156	7,738	**−4.50 (−0.59, 426)**	−5.52 (−1.52, 425)	**−4.50 (−0.59, 426)**	**−4.50 (−0.59, 426)**
TS1-NA-IA-50-I2	2,779	10,883	**−0.28 (−0.27, 455)**	−0.60 (−0.50, 467)	−0.50 (−0.50, 467)	−0.50 (−0.50, 472)
TS1-NA-IA-50-I3	2,779	11,019	**−0.02 (−0.02, 458)**	**−0.02 (−0.02, 458)**	**−0.02 (−0.02, 458)**	**−0.02 (−0.02, 458)**
TS1-NA-IA-50-I4	2,751	10,178	**−4.97 (−2.05, 384)**	−8.24 (−3.80, 390)	−8.30 (−3.80, 383)	−8.23 (−3.80, 387)
TS1-NA-IA-50-I5	2,779	10,747	**−5.63 (−1.94, 515)**	−5.78 (−1.94, 516)	**−5.63 (−1.94, 515)**	**−5.63 (−1.94, 515)**
TS1-NA-MUT-25-I1	2,182	8,033	**−1.08 (−0.46, 544)**	**−1.08 (−0.46, 544)**	**−1.08 (−0.46, 544)**	**−1.08 (−0.46, 544)**
TS1-NA-MUT-25-I2	2,779	10,611	**−1.65 (−0.86, 470)**	−2.29 (−0.82, 469)	−2.29 (−0.82, 470)	−1.97 (−0.77, 442)
TS1-NA-MUT-25-I3	2,687	10,881	**0.00 (0.33 , 482)**	0.00 (0.33, 481)	**0.00 (0.33 , 482)**	**0.00 (0.33 , 482)**
TS1-NA-MUT-25-I4	2,779	10,883	**−0.22 (−0.22, 502)**	−0.40 (−0.22, 492)	−0.40 (−0.22, 495)	−0.40 (−0.22, 498)
TS1-NA-MUT-25-I5	2,779	10,611	**−1.91 (−1.18, 433)**	−2.89 (−0.68, 429)	−2.52 (−0.68, 431)	**−1.91 (−1.18, 433)**
TS1-NA-MUT-50-I1	2,779	11,019	**−0.27 (−0.27, 474)**	**−0.27 (−0.27, 474)**	**−0.27 (−0.27, 474)**	**−0.27 (−0.27, 474)**
TS1-NA-MUT-50-I2	2,182	8,033	**−4.34 (−0.92, 556)**	**−4.34 (−0.92, 556)**	**−4.34 (−0.92, 556)**	**−4.34 (−0.92, 556)**
TS1-NA-MUT-50-I3	2,779	11,019	**−1.66 (−0.77, 453)**	−1.66 (−0.77, 452)	**−1.66 (−0.77, 453)**	**−1.66 (−0.77, 453)**
TS1-NA-MUT-50-I4	2,182	8,101	**−0.90 (−0.88, 529)**	**−0.90 (−0.88, 529)**	**−0.90 (−0.88, 529)**	**−0.90 (−0.88, 529)**
TS1-NA-MUT-50-I5	2,779	10,747	**−3.64 (−2.91, 451)**	−5.79 (−1.57, 444)	**−3.64 (−2.91, 451)**	**−3.64 (−2.91, 451)**
TS1-UA-IA-25-I1	2,182	8,033	**−0.89 (−0.53, 544)**	**−0.89 (−0.53, 544)**	**−0.89 (−0.53, 544)**	**−0.89 (−0.53, 544)**
TS1-UA-IA-25-I2	2,719	9,972	**−1.83 (−0.75, 341)**	−2.70 (−1.20, 428)	−2.51 (−0.87, 428)	−2.33 (−0.87, 429)
TS1-UA-IA-25-I3	2,779	10,883	**−0.39 (−0.39, 519)**	−0.47 (−0.47, 509)	**−0.39 (−0.39, 519)**	**−0.39 (−0.39, 519)**
TS1-UA-IA-25-I4	2,779	11,019	**−0.20 (−0.19, 460)**	−0.46 (−0.33, 466)	−0.24 (−0.11, 454)	−0.24 (−0.11, 456)
TS1-UA-IA-25-I5	2,751	10,450	**−0.78 (−0.23, 291)**	−3.02 (−1.12, 386)	−2.68 (−0.85, 390)	−2.48 (−0.63, 347)
TS1-UA-IA-50-I1	2,779	11,019	**−1.26 (−0.77, 474)**	−4.68 (−1.37, 501)	**−1.26 (−0.77, 474)**	**−1.26 (−0.77, 474)**
TS1-UA-IA-50-I2	2,779	10,475	**−5.18 (−2.79, 390)**	−5.35 (−2.79, 387)	**−5.18 (−2.79, 390)**	**−5.18 (−2.79, 390)**
TS1-UA-IA-50-I3	2,156	7,806	**−0.95 (−0.43, 430)**	−3.52 (−1.41, 431)	**−0.95 (−0.43, 430)**	**−0.95 (−0.43, 430)**
TS1-UA-IA-50-I4	2,727	10,360	**−2.57 (−1.59, 449)**	−3.39 (−2.03, 454)	−2.70 (−1.59, 453)	−2.61 (−1.59, 449)
TS1-UA-IA-50-I5	2,779	10,883	**−2.59 (−0.78, 468)**	−2.81 (−1.00, 464)	−2.81 (−1.00, 464)	−2.81 (−1.00, 464)
TS1-UA-MUT-25-I1	2,182	8,033	**−0.71 (−0.35, 544)**	**−0.71 (−0.35, 544)**	**−0.71 (−0.35, 544)**	**−0.71 (−0.35, 544)**
TS1-UA-MUT-25-I2	2,779	10,611	**−1.92 (−0.48, 470)**	−2.33 (−0.73, 469)	**−1.92 (−0.48, 470)**	**−1.92 (−0.48, 470)**
TS1-UA-MUT-25-I3	2,687	10,881	**0.00 (0.26 , 482)**	0.00 (0.26, 481)	**0.00 (0.26 , 482)**	**0.00 (0.26 , 482)**
TS1-UA-MUT-25-I4	2,779	10,883	**−0.32 (−0.32, 503)**	−0.32 (−0.32, 496)	**−0.32 (−0.32, 503)**	**−0.32 (−0.32, 503)**
TS1-UA-MUT-25-I5	2,779	10,611	**−1.63 (−0.85, 443)**	−3.02 (−1.01, 429)	−3.02 (−1.01, 430)	**−1.63 (−0.85, 433)**
TS1-UA-MUT-50-I1	2,779	11,019	**−1.00 (−0.55, 474)**	−3.56 (−1.27, 500)	**−1.00 (−0.55, 474)**	**−1.00 (−0.55, 474)**
TS1-UA-MUT-50-I2	2,182	8,033	**−1.77 (−1.02, 557)**	**−1.77 (−1.02, 557)**	**−1.77 (−1.02, 557)**	**−1.77 (−1.02, 557)**
TS1-UA-MUT-50-I3	2,779	11,019	**−1.22 (−0.59, 453)**	−1.22 (−0.59, 452)	−1.22 (−0.59, 452)	−1.22 (−0.59, 452)
TS1-UA-MUT-50-I4	2,182	8,101	**−2.24 (−0.68, 528)**	−2.34 (−0.80, 528)	**−2.24 (−0.68, 528)**	**−2.24 (−0.68, 528)**
TS1-UA-MUT-50-I5	2,779	10,747	**−2.98 (−2.24, 453)**	−4.48 (−2.24, 446)	−4.93 (−2.24, 442)	−4.48 (−2.24, 446)
TS2-NA-IA-25-I1	5,723	22,020	**−0.61 (−0.32, 688)**	−1.59 (−0.65, 679)	−0.98 (−0.65, 680)	−0.98 (−0.65, 680)
TS2-NA-IA-25-I2	4,820	19,085	**0.00 (0.26 , 803)**	0.00 (0.20, 794)	0.00 (0.26, 799)	0.00 (0.26, 800)
TS2-NA-IA-25-I3	5,586	22,018	**0.00 (0.05 , 717)**	−0.15 (−0.15, 713)	−0.15 (−0.15, 720)	−0.15 (−0.15, 721)
TS2-NA-IA-25-I4	5,525	19,659	**−0.74 (−0.36, 537)**	−3.26 (−0.78, 557)	−1.65 (−1.07, 553)	−0.98 (−0.36, 582)
TS2-NA-IA-25-I5	5,586	21,814	**0.00 (0.03 , 652)**	−1.30 (−0.52, 603)	−0.65 (−0.52, 662)	−0.64 (−0.36, 628)
TS2-NA-IA-50-I1	5,645	21,535	**−1.41 (−0.50, 669)**	−1.41 (−0.50, 668)	−1.57 (−0.61, 667)	−1.41 (−0.50, 668)
TS2-NA-IA-50-I4	5,621	21,261	**−1.76 (−0.82, 757)**	−2.71 (−0.82, 763)	−1.76 (−0.82, 756)	−1.76 (−0.82, 756)
TS2-NA-MUT-25-I1	5,723	22,632	**−0.53 (−0.22, 692)**	−0.86 (−0.43, 676)	−0.86 (−0.43, 677)	−0.86 (−0.43, 677)
TS2-NA-MUT-25-I2	4,957	18,951	**0.00 (0.29 , 803)**	−0.29 (−0.29, 805)	0.00 (0.20, 805)	0.00 (0.20, 806)
TS2-NA-MUT-25-I3	5,723	22,632	**−0.22 (−0.22, 699)**	−0.53 (−0.44, 692)	−0.53 (−0.44, 693)	−0.22 (−0.22, 692)
TS2-NA-MUT-25-I4	5,723	22,020	**−0.26 (−0.21, 670)**	−0.95 (−0.54, 668)	−0.31 (−0.21, 666)	−0.26 (−0.21, 664)
TS2-NA-MUT-25-I5	5,586	22,630	**−0.18 (−0.05, 601)**	−0.93 (−0.37, 605)	−0.47 (−0.19, 594)	−0.38 (−0.19, 600)
TS2-NA-MUT-50-I4	5,723	22,020	**−2.37 (−0.76, 726)**	−4.18 (−1.87, 735)	−3.93 (−1.87, 734)	−2.69 (−0.76, 725)
TS2-UA-IA-25-I2	4,885	17,911	**0.00 (0.32 , 802)**	0.00 (0.16, 795)	0.00 (0.24, 800)	**0.00 (0.32 , 802)**
TS2-UA-IA-25-I3	5,675	21,676	**−0.20 (−0.20, 719)**	−0.41 (−0.20, 707)	−0.20 (−0.20, 718)	−0.20 (−0.20, 718)
TS2-UA-IA-25-I5	4,957	18,679	**−0.35 (−0.17, 632)**	−0.87 (−0.69, 610)	−0.87 (−0.69, 626)	−0.69 (−0.20, 618)
TS2-UA-IA-50-I1	4,820	19,085	**−1.98 (−0.67, 657)**	−3.08 (−0.60, 654)	−2.69 (−0.68, 646)	−2.14 (−0.83, 655)
TS2-UA-IA-50-I2	5,723	22,020	**−0.59 (−0.36, 587)**	−2.00 (−0.59, 586)	−0.68 (−0.36, 586)	−0.68 (−0.36, 586)
TS2-UA-MUT-25-I2	5,723	21,816	**0.00 (0.21 , 807)**	0.00 (0.21, 798)	0.00 (0.21, 802)	0.00 (0.21, 806)
TS2-UA-MUT-25-I3	5,645	21,535	**−0.29 (−0.29, 702)**	−0.48 (−0.29, 694)	−0.29 (−0.29, 698)	−0.29 (−0.29, 701)
TS2-UA-MUT-25-I4	5,621	21,261	**−0.27 (−0.27, 675)**	−0.60 (−0.32, 653)	−0.27 (−0.27, 652)	−0.27 (−0.27, 668)
TS3-NA-IA-25-I3	5,586	22,630	**0.00 (0.60 , 988)**	0.00 (0.28, 946)	0.00 (0.49, 946)	0.00 (0.52, 948)
			Average deviation from optimal	−0.68 (−0.20, 2.86)	−0.31 (−0.12, 3.42)	−0.19 (−0.07, 2.84)
			Standard deviation from optimal	0.88 (0.41, 24.20)	0.60 (0.31, 22.30)	0.50 (0.24, 20.60)

produces solutions with lower WO values in several cases, particularly for the largest test scenarios, while the opposite is true for the Greedy approach. While It-RG-SA outperforms It-RG(100) in all cases, total water output is lower for some cases when running more iterations of It-RG-SA, which comes at a trade-off with improved water target completion values.

To gain insight into the structure of the obtained schedules, the distribution of resource types allocated to each area over time was also evaluated.[4] As expected, most areas have a single resource type allocated to them for most (or all) time slots. This makes sense since only resources of the same type can be active simultaneously in the same area. Consequently, maintaining the same type over the course of

[4] This data was post-processed from optimal MILP solutions and It-RG-SA(150) solutions for all test scenarios. All data can be found at https://github.com/LMesaric/Aerial-Resource-Scheduler.

Table 12
Results obtained for instances with $CF = 25\%$, given as TD (TC_{min}, WO) [kl].

TS	Target Dist.	MILP	Greedy	RG	It-RG(100)	It-RG-SA(1)	It-RG-SA(20)	It-RG-SA(150)
TS1	NA-IA	**-0.82** (**-0.33**, **437**)	−3.81 (−0.88, 454)	−5.52 (−1.19, 453)	−2.00 (−0.62, 462)	−1.48 (−0.51, 470)	−1.16 (−0.46, 469)	−0.99 (−0.40, 467)
TS1	NA-MUT	**-0.97** (**-0.48**, **486**)	−5.88 (−1.06, 473)	−8.28 (−1.59, 462)	−3.39 (−1.11, 470)	−1.33 (−0.37, 483)	−1.26 (−0.37, 484)	−1.07 (−0.46, 480)
TS1	UA-IA	**-0.82** (**-0.42**, **431**)	−3.12 (−0.76, 455)	−6.07 (−1.20, 447)	−1.88 (−0.70, 465)	−1.51 (−0.73, 467)	−1.34 (−0.55, 467)	−1.26 (−0.50, 459)
TS1	UA-MUT	**-0.92** (**-0.35**, **488**)	−3.65 (−0.71, 482)	−5.43 (−1.09, 469)	−2.24 (−0.69, 482)	−1.28 (−0.43, 484)	−1.19 (−0.38, 486)	−0.92 (−0.35, 486)
TS2	NA-IA	**-0.27** (**-0.07**, **679**)	−4.22 (−0.84, 670)	−6.28 (−1.12, 657)	−2.67 (−0.94, 674)	−1.26 (−0.38, 669)	−0.69 (−0.42, 683)	−0.55 (−0.25, 682)
TS2	NA-MUT	**-0.24** (**-0.08**, **693**)	−4.38 (−1.18, 677)	−9.63 (−1.44, 647)	−2.78 (−0.68, 680)	−0.71 (−0.41, 689)	−0.43 (−0.21, 687)	−0.34 (−0.17, 688)
TS2	UA-IA	**-0.47** (**-0.19**, **673**)	−4.52 (−1.07, 669)	−8.61 (−1.51, 659)	−2.89 (−0.92, 659)	−1.22 (−0.54, 674)	−0.82 (−0.48, 678)	−0.74 (−0.35, 669)
TS2	UA-MUT	**-0.29** (**-0.18**, **701**)	−4.57 (−1.06, 681)	−7.23 (−1.13, 672)	−2.68 (−0.90, 677)	−0.52 (−0.19, 683)	−0.34 (−0.18, 688)	−0.34 (−0.18, 696)
TS3	NA-IA	**0.00** (**0.38**, **948**)	−2.61 (−0.89, 972)	−6.49 (−1.92, 952)	−0.58 (−0.33, 969)	−0.19 (0.06, 989)	−0.09 (0.26, 984)	−0.05 (0.27, 987)
TS3	NA-MUT	**0.00** (**0.33**, **885**)	−3.00 (−1.07, 911)	−6.25 (−1.43, 885)	−1.42 (−0.66, 901)	−0.26 (−0.18, 920)	−0.10 (0.19, 920)	−0.01 (0.35, 919)
TS3	UA-IA	**0.00** (**0.31**, **886**)	−2.03 (−0.75, 947)	−4.70 (−1.42, 924)	−0.81 (−0.47, 920)	−0.21 (−0.08, 961)	−0.08 (0.12, 958)	−0.05 (0.23, 944)
TS3	UA-MUT	−0.08 (0.19, 927)	−4.47 (−1.44, 895)	−6.32 (−1.55, 880)	−1.09 (−0.43, 903)	−0.30 (−0.25, 931)	−0.08 (0.21, 921)	**-0.08** (**0.24**, **919**)
TS4	NA-IA	0.00 (0.34, 1,287)	−2.93 (−1.17, 1,274)	−5.25 (−2.19, 1,294)	−0.44 (−0.26, 1,278)	−0.24 (0.06, 1,329)	−0.07 (0.32, 1,320)	**0.00** (**0.57**, **1312**)
TS4	NA-MUT	**-0.04** (**-0.01**, **1314**)	−3.75 (−0.82, 1,255)	−4.73 (−1.22, 1,212)	−0.96 (−0.42, 1,240)	−0.32 (−0.09, 1,286)	−0.12 (0.17, 1,277)	−0.12 (0.23, 1,280)
TS4	UA-IA	**0.00** (**0.38**, **1314**)	−1.73 (−0.87, 1,308)	−6.34 (−1.73, 1,250)	−0.80 (−0.38, 1,257)	−0.27 (0.10, 1,304)	−0.01 (0.50, 1,330)	−0.01 (0.61, 1,333)
TS4	UA-MUT	**-0.09** (**0.05**, **1241**)	−4.95 (−1.31, 1,286)	−8.25 (−1.31, 1,219)	−1.52 (−0.42, 1,230)	−0.49 (0.02, 1,288)	−0.18 (0.13, 1,309)	−0.22 (0.13, 1,314)
TS5	NA-IA	0.00 (0.11, 1,602)	−2.08 (−0.91, 1,603)	−4.36 (−1.66, 1,558)	−0.43 (−0.29, 1,545)	−0.23 (0.24, 1,641)	−0.04 (0.58, 1,629)	**0.00** (**0.68**, **1572**)
TS5	NA-MUT	0.00 (0.10, 1,569)	−3.68 (−1.48, 1,582)	−4.05 (−1.47, 1,544)	−0.56 (−0.29, 1,589)	−0.36 (−0.14, 1,624)	−0.18 (0.16, 1,630)	**0.00** (**0.41**, **1605**)
TS5	UA-IA	0.00 (0.18, 1,537)	−2.04 (−0.95, 1,630)	−5.57 (−1.70, 1,601)	−0.30 (−0.18, 1,565)	−0.15 (0.60, 1,585)	−0.07 (0.88, 1,624)	**0.00** (**1.01**, **1622**)
TS5	UA-MUT	**0.00** (**0.15**, **1566**)	−4.44 (−1.30, 1,647)	−5.74 (−1.73, 1,617)	−0.49 (−0.17, 1,588)	−0.13 (0.21, 1,648)	−0.10 (0.59, 1,659)	−0.10 (0.63, 1,658)
TS6	NA-IA	0.00 (0.11, 2,016)	−0.71 (−0.43, 1,907)	−4.27 (−1.87, 1,880)	−0.77 (−0.27, 1,827)	0.00 (0.72, 1,925)	0.00 (0.98, 1,921)	**0.00** (**1.08**, **1910**)
TS6	NA-MUT	0.00 (0.07, 1,927)	−3.87 (−1.81, 1,862)	−7.49 (−1.95, 1,796)	−1.13 (−0.57, 1,828)	0.00 (0.40, 1,846)	0.00 (0.95, 1,927)	**0.00** (**0.99**, **1930**)
TS6	UA-IA	0.00 (0.12, 1,936)	−3.24 (−0.94, 1,944)	−6.14 (−1.54, 1,882)	−0.65 (−0.30, 1,881)	0.00 (0.45, 1,960)	0.00 (0.84, 1,938)	**0.00** (**0.99**, **1927**)
TS6	UA-MUT	0.00 (0.08, 1,972)	−4.79 (−1.08, 1,898)	−6.25 (−1.81, 1,825)	−0.84 (−0.39, 1,778)	0.00 (0.47, 1,895)	0.00 (0.84, 1,888)	**0.00** (**0.88**, **1896**)
TS7	NA-IA	0.00 (0.03, 1,887)	−1.58 (−0.58, 2,070)	−7.04 (−2.43, 2,047)	−0.73 (−0.30, 2,032)	−0.23 (0.13, 2,080)	−0.02 (0.59, 2,095)	**0.00** (**0.75**, **2051**)
TS7	NA-MUT	**0.00** (**0.06**, **2149**)	−3.95 (−1.42, 2,118)	−7.08 (−2.06, 2,094)	−0.70 (−0.38, 2,063)	−0.28 (0.23, 2,164)	−0.05 (0.70, 2,105)	−0.05 (0.73, 2,136)
TS7	UA-IA	−0.01 (0.07, 2,257)	−5.86 (−1.13, 2,410)	−8.06 (−2.38, 2,278)	−1.16 (−0.44, 2,282)	−0.10 (0.47, 2,374)	0.00 (0.83, 2,352)	**0.00** (**0.92**, **2421**)
TS7	UA-MUT	0.00 (0.07, 2,326)	−2.60 (−0.87, 2,324)	−2.96 (−1.06, 2,243)	−1.01 (−0.54, 2,208)	−0.11 (0.44, 2,281)	0.00 (0.92, 2,283)	**0.00** (**0.99**, **2324**)
TS8	NA-IA	−48.06 (−4.28, 2,555)	−35.68 (−5.29, 3,038)	−94.19 (−7.27, 2,999)	−39.58 (−6.01, 2,986)	−9.13 (−1.97, 3,141)	−2.42 (−1.09, 3,087)	**-1.34** (**-0.52**, **3086**)
TS8	NA-MUT	−18.90 (−2.45, 2,634)	−27.87 (−4.58, 3,121)	−61.33 (−6.26, 3,056)	−22.40 (−3.19, 3,018)	−3.39 (−1.83, 3,129)	−0.25 (−0.13, 3,166)	**-0.12** (**-0.01**, **3130**)
TS8	UA-IA	−3.24 (−0.84, 2,676)	−5.26 (−1.96, 3,264)	−12.85 (−2.14, 3,075)	−4.07 (−0.87, 3,072)	−0.15 (0.22, 3,139)	−0.06 (0.41, 3,215)	**0.00** (**0.58**, **3129**)
TS8	UA-MUT	−3.28 (−0.77, 2,713)	−6.76 (−1.47, 3,274)	−10.08 (−1.34, 3,038)	−3.67 (−0.84, 3,023)	−0.21 (−0.03, 3,277)	0.00 (0.52, 3,198)	**0.00** (**0.59**, **3192**)

Table 13
Results obtained for instances with $CF = 50\%$, given as TD (TC_{min}, WO) [kl].

TS	Target Dist.	MILP	Greedy	RG	It-RG(100)	It-RG-SA(1)	It-RG-SA(20)	It-RG-SA(150)
TS1	NA-IA	**−3.08** (**−0.97**, **448**)	−12.34 (−2.56, 450)	−17.57 (−3.34, 443)	−6.90 (−1.63, 452)	−4.03 (−1.56, 451)	−3.79 (−1.37, 450)	−3.77 (−1.37, 452)
TS1	NA-MUT	**−2.16** (**−1.15**, **493**)	−14.27 (−3.05, 490)	−20.32 (−3.14, 465)	−4.50 (−1.14, 488)	−2.59 (−0.88, 491)	−2.16 (−1.15, 493)	**−2.16** (**−1.15**, **493**)
TS1	UA-IA	**−2.51** (**−1.27**, **442**)	−9.81 (−1.77, 446)	−18.60 (−3.05, 430)	−5.53 (−2.04, 439)	−3.95 (−1.72, 447)	−2.58 (−1.32, 442)	−2.56 (−1.32, 441)
TS1	UA-MUT	**−1.84** (**−1.01**, **493**)	−11.98 (−2.43, 494)	−14.04 (−2.55, 480)	−6.66 (−1.26, 487)	−2.67 (−1.18, 497)	−2.23 (−1.01, 491)	−2.14 (−1.01, 492)
TS2	NA-IA	**−0.88** (**−0.42**, **722**)	−13.21 (−2.12, 712)	−20.08 (−2.90, 701)	−6.06 (−1.57, 709)	−2.20 (−0.66, 718)	−1.03 (−0.47, 717)	−0.89 (−0.42, 721)
TS2	NA-MUT	**−2.80** (**−0.86**, **653**)	−16.45 (−2.68, 655)	−17.86 (−2.62, 651)	−9.46 (−1.87, 649)	−5.30 (−1.88, 655)	−3.20 (−1.39, 656)	−2.95 (−1.18, 655)
TS2	UA-IA	**−1.43** (**−0.57**, **712**)	−13.20 (−2.46, 712)	−21.91 (−3.38, 692)	−7.76 (−1.42, 692)	−2.66 (−0.86, 705)	−1.67 (−0.57, 710)	−1.56 (−0.60, 713)
TS2	UA-MUT	**−4.55** (**−1.70**, **648**)	−20.59 (−2.02, 659)	−31.84 (−2.16, 642)	−14.86 (−2.10, 642)	−6.20 (−1.61, 651)	−5.08 (−1.70, 645)	−4.76 (−1.70, 647)
TS3	NA-IA	**−2.55** (**−1.36**, **932**)	−12.06 (−3.10, 983)	−29.62 (−5.04, 960)	−10.50 (−3.05, 985)	−6.78 (−3.06, 985)	−3.46 (−1.72, 974)	−3.08 (−1.49, 979)
TS3	NA-MUT	−2.76 (−1.34, 907)	−15.18 (−3.29, 896)	−15.61 (−3.03, 895)	−8.24 (−2.46, 891)	−3.19 (−1.50, 896)	−2.79 (−1.41, 914)	**−2.65** (**−1.30**, **905**)
TS3	UA-IA	**−3.44** (**−1.45**, **933**)	−15.31 (−3.57, 978)	−22.20 (−4.52, 958)	−8.90 (−2.73, 964)	−7.09 (−2.55, 981)	−3.88 (−1.45, 976)	−3.54 (−1.45, 978)
TS3	UA-MUT	−4.19 (−1.33, 889)	−15.92 (−2.85, 903)	−23.29 (−2.99, 879)	−11.89 (−2.68, 891)	−6.18 (−1.86, 901)	−4.16 (−1.65, 894)	**−3.67** (**−1.42**, **892**)
TS4	NA-IA	−2.01 (−1.31, 1,190)	−14.76 (−2.99, 1,205)	−26.95 (−4.94, 1,191)	−11.39 (−2.94, 1,174)	−4.61 (−1.74, 1,216)	−1.80 (−1.30, 1,217)	**−1.51** (**−0.67**, **1218**)
TS4	NA-MUT	−2.86 (−1.33, 1,220)	−14.83 (−3.54, 1,267)	−20.35 (−4.03, 1,236)	−8.93 (−3.05, 1,267)	−3.55 (−1.56, 1,261)	−2.49 (−1.25, 1,279)	**−2.34** (**−1.14**, **1283**)
TS4	UA-IA	−2.38 (−0.94, 1,228)	−10.87 (−3.41, 1,247)	−27.98 (−4.28, 1,221)	−8.28 (−2.47, 1,216)	−2.66 (−0.91, 1,239)	−1.90 (−0.89, 1,253)	**−1.23** (**−0.25**, **1247**)
TS4	UA-MUT	−4.31 (−2.01, 1,202)	−20.71 (−3.04, 1,271)	−26.46 (−3.11, 1,204)	−12.37 (−2.46, 1,230)	−4.85 (−1.74, 1,272)	−2.67 (−1.08, 1,271)	**−1.79** (**−0.72**, **1257**)
TS5	NA-IA	−0.87 (−0.26, 1,612)	−10.41 (−3.14, 1,589)	−28.25 (−6.01, 1,604)	−11.37 (−3.58, 1,570)	−2.58 (−1.09, 1,627)	−0.39 (−0.12, 1,606)	**−0.38** (**−0.09**, **1628**)
TS5	NA-MUT	−0.31 (−0.25, 1,524)	−12.77 (−4.28, 1,603)	−29.53 (−5.05, 1,593)	−5.20 (−1.39, 1,595)	−0.75 (−0.52, 1,598)	−0.15 (0.14, 1,629)	**−0.11** (**0.17**, **1648**)
TS5	UA-IA	−0.49 (−0.20, 1,671)	−9.96 (−2.86, 1,681)	−35.25 (−4.86, 1,657)	−12.32 (−2.44, 1,624)	−9.71 (−2.58, 1,689)	−0.32 (0.06, 1,671)	**−0.14** (**0.22**, **1682**)
TS5	UA-MUT	−0.21 (−0.16, 1,629)	−17.35 (−3.64, 1,634)	−21.51 (−3.66, 1,593)	−6.71 (−2.32, 1,585)	−1.23 (−0.49, 1,656)	−0.06 (0.17, 1,633)	**−0.06** (**0.24**, **1644**)
TS6	NA-IA	−1.56 (−0.58, 1,857)	−20.83 (−3.83, 1,977)	−62.63 (−7.48, 1,930)	−12.89 (−3.48, 1,908)	−2.89 (−1.05, 1,985)	−0.48 (0.00, 1,972)	**−0.32** (**0.16**, **1966**)
TS6	NA-MUT	−1.26 (−0.50, 1,900)	−19.57 (−3.70, 1,924)	−36.34 (−4.99, 1,924)	−11.31 (−2.41, 1,881)	−0.93 (−0.40, 1,912)	**−0.28** (**0.05**, **1959**)	−0.28 (0.04, 1,946)
TS6	UA-IA	−1.18 (−0.59, 1,842)	−35.69 (−5.48, 1,978)	−67.11 (−4.82, 1,917)	−22.42 (−4.11, 1,957)	−7.59 (−1.55, 1,962)	−0.52 (−0.32, 1,987)	**−0.17** (**−0.02**, **1984**)
TS6	UA-MUT	−2.96 (−0.89, 1,988)	−17.95 (−2.93, 2,046)	−34.00 (−3.38, 1,962)	−11.89 (−2.82, 1,994)	−8.49 (−1.23, 2,051)	−0.58 (−0.18, 2,085)	**−0.40** (**−0.10**, **2057**)
TS7	NA-IA	−3.93 (−2.00, 1,926)	−31.81 (−6.15, 2,151)	−83.78 (−8.32, 2,112)	−27.51 (−7.50, 2,076)	−7.96 (−1.73, 2,152)	−0.86 (−0.52, 2,172)	**−0.61** (**−0.39**, **2167**)
TS7	NA-MUT	−3.69 (−1.43, 1,974)	−29.75 (−3.56, 2,160)	−52.60 (−6.31, 2,116)	−16.73 (−3.51, 2,120)	−3.08 (−1.66, 2,168)	−0.33 (−0.08, 2,192)	**−0.10** (**0.22**, **2177**)
TS7	UA-IA	−4.54 (−1.50, 2,100)	−54.14 (−5.17, 2,333)	−54.16 (−5.44, 2,198)	−28.97 (−4.58, 2,201)	−14.23 (−3.18, 2,306)	−0.15 (0.00, 2,270)	**−0.15** (**−0.02**, **2239**)
TS7	UA-MUT	−0.85 (−0.37, 2,114)	−33.96 (−3.90, 2,207)	−36.81 (−4.17, 2,141)	−9.23 (−2.73, 2,125)	−4.06 (−1.04, 2,185)	−0.37 (−0.10, 2,210)	**−0.20** (**0.10**, **2192**)
TS8	NA-IA	−441.04 (−18.68, 2,681)	−532.59 (−17.61, 3,105)	−776.93 (−21.34, 3,039)	−489.44 (−22.44, 3,006)	−318.93 (−14.90, 3,020)	−218.33 (−13.81, 3,001)	**−187.93** (**−13.52**, **3012**)
TS8	NA-MUT	−1179.20 (−16.26, 2,632)	−475.20 (−14.34, 3,172)	−649.21 (−16.18, 3,089)	−321.56 (−14.49, 3,042)	−188.34 (−11.47, 3,052)	−100.62 (−7.96, 3,056)	**−74.09** (**−8.50**, **3048**)
TS8	UA-IA	−42.89 (−4.69, 2,612)	−72.76 (−5.33, 3,212)	−127.97 (−5.80, 3,026)	−64.87 (−5.00, 3,048)	−8.79 (−1.61, 3,140)	−3.08 (−0.74, 3,183)	**−1.54** (**−0.51**, **3132**)
TS8	UA-MUT	−41.76 (−3.65, 2,543)	−87.52 (−3.95, 3,236)	−77.37 (−3.62, 3,057)	−48.21 (−3.63, 2,994)	−3.64 (−0.79, 3,116)	−1.28 (−0.53, 3,104)	**−1.04** (**−0.52**, **3048**)

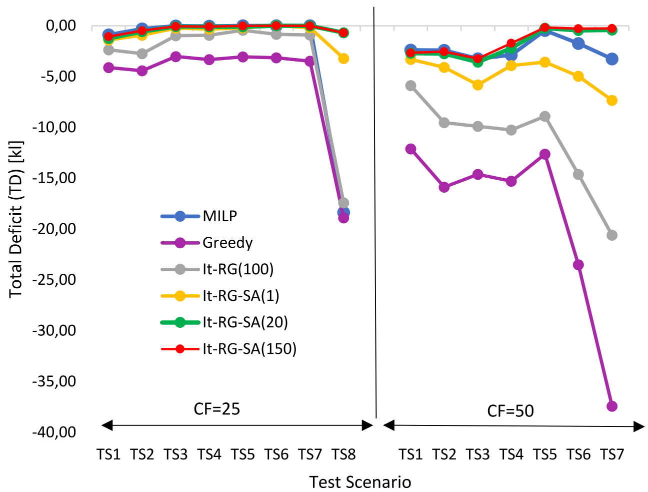

Fig. 2. Average Total Deficit (TD) for each TS and CF.

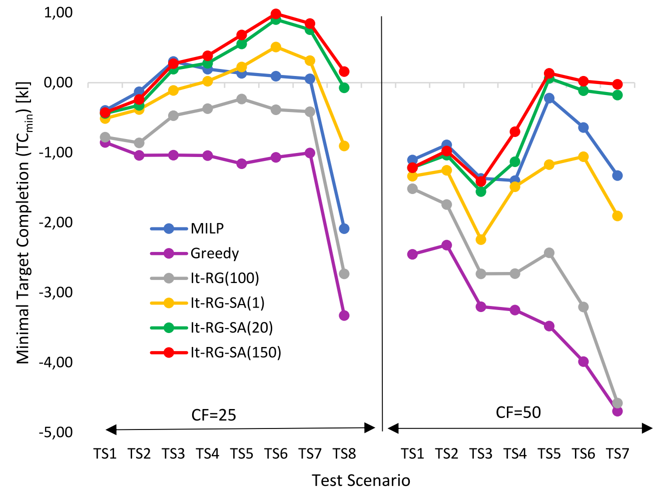

Fig. 3. Average Minimal Target Completion (TC_{min}) for each TS and CF.

Table 14
Average execution time [s].

TS	MILP	RG phase	SA phase	It-RG-SA(1)
TS1	195.839	0.006	3.887	3.893
TS2	3,841.898*	0.040	7.571	7.611
TS3	1,870.504*	0.118	7.950	8.068
TS4	-	0.565	12.640	13.205
TS5	-	1.122	13.390	14.512
TS6	-	3.189	20.505	23.694
TS7	-	4.919	21.065	25.983
TS8	-	41.913	50.136	92.049

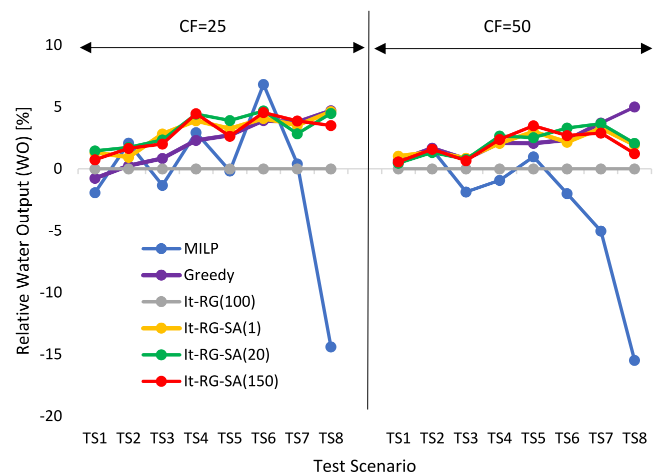

Fig. 4. Average Relative Water Output (WO) for each TS and CF.

the day allows for easier scheduling. For smaller scenarios (TS1-TS3), 45 %–65 % of areas had a single type allocated to them during all time slots, while for larger scenarios (TS4-TS8) this occurred for 70 %–80 % of areas.

Table 14 shows the average execution time required for each test scenario. Note, the average execution time shown for the MILP is only for cases where the optimal solution was found. While this occurs for all problem instances in TS1, for TS2 and TS3 the MILP found optimal solutions in only 21 and 1 case out of 40, respectively, while the remaining instances timed out. Consequently, the values shown are not generally applicable for that problem size, particularly for TS3, and are marked with '*'. The average execution time for the heuristic is shown per It-RG-SA iteration, measured in seconds and broken down into the individual RG and SA phases. The arithmetic means for It-RG-SA were calculated according to 6 000 iterations for each test scenario size, with deviation of execution time of individual iterations below 11 % for all sizes. Different water target distributions and CF values did not give noticeable differences in execution time. Note, while execution time grows linearly with the number of It-RG-SA iterations, individual iterations are independent, allowing for easy parallelization. For example, actual tests for It-RG-SA utilized 4 cores so elapsed real times for the largest TS8 instances were ~8 min (20 iterations) and ~60 min (150 iterations). If 16 cores are used, these times could be reduced to ~2 min (20 iterations) and ~15 min (150 iterations). Depending on the planning phase, whether scheduling the night before allowing for an approximate 2 h time window, or performing more time-sensitive early morning scheduling allowing for approximately 10 to 15 min, the algorithm could be run for an appropriate number of It-RG-SA iterations. Also note that individual SA phases were run for $L = 10^4$ iterations each. In case of stricter time constraints, the number of SA iterations could be reduced presumably without significant degradation of results (see Fig. 1).

6. Conclusions

The increasing number of aerial resources being mobilized to aid in the containment of extreme wildfire events in recent years calls for more efficient resource allocation and scheduling algorithms. In this paper, we present a MILP optimization model for the daily scheduling of aerial resource operations for the suppression of large-scale wildfires. The main objective is to achieve a given target water flow in each area of operation and time period, to ensure prioritized non-negligence of all fire areas over all daylight hours. For larger instances, an efficient GRASP-inspired multi-start heuristic algorithm is proposed based on a randomized greedy construction phase, coupled with simulated annealing and large neighborhood search techniques. A diverse set of problem instances is generated based on realistic data to test the approaches and serve as a benchmark for future work. Results indicate that the MILP model can be solved to optimality for instances with up to 15 resources and 3 areas. The heuristic was shown to find optimal or near-optimal solutions for these instances, and gives solutions approaching target water flows for larger problem sizes. The algorithm allows for easy parallelization and was shown to give good results in a small number of iterations, making it applicable not only for night-before planning, but also early-morning scheduling in a dynamic firefighting environment.

In future work, we aim to develop a dynamic re-scheduling algorithm which departs from a currently ongoing schedule, but adjusts it over the course of the day to efficiently adapt to unexpected variations. An additional avenue of research is focused on extending the approach to include initial resource allocation and extinction strategy decisions which could aid in the development of more efficient automated resource allocation tools. Additional objectives could be considered, such as minimizing burned area or cost, and multi-criteria optimization techniques, such as Multi-Criteria Decision Making (MCDM) or Goal Programming (GP), could be applied to investigate the associated trade-offs between these objectives.

CRediT authorship contribution statement

Nina Skorin-Kapov: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Luka Mesarić:** Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Fernando Pereñíguez García:** Software, Visualization. **Lea Skorin-Kapov:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

There are no conflicts of interest.

Data availability

A link to all data and code is included in the manuscript (publically available on Github).

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